# THE DROPPED $m = 0$ COEFFICIENT AT A SPLIT LEVEL PRIME

**Status.** This is the argument for Proposition 4 below and for the $m = 0$ correction it implies, extracted so it can be read on its own. It has been reviewed twice by agents and corrected both times; it is *not* yet agreed, and nothing in the accompanying code or tests describes it as proved. Objects and notation ($L_-$, $L_+$, $\lambda$, the Eisenstein coefficients $A_\mu(s, m, v)$, the multiplier of Observation obs:mult) are those of the preprint's Sections sec:chain and sec:data; cross-references into the preprint are left as plain labels.

**The literature's zero, and where it comes from.** In the published form of Schofer's theorem [1, Thm. 1.1] the sum runs over all cosets $\eta$ and all $m \geq 0$; the vanishing of the $m = 0$ terms at $\eta \neq 0$ is [1, Lemma 2.21], "for $\mu \neq 0$, $b_\mu(0, t) = 0$", stated there for a general lattice. Its proof reads: "Theorem 5.2 of [17] implies $W_{0,p}(s, \varphi_{\mu,p}) = 0$ if $\varphi_{\mu,p}$ is not the characteristic function of the local lattice", where [17] is an unpublished preprint (T. Yang, 2001), so the step cannot be checked as stated. The nearest published statement we can check is [3, Prop. 5.2], and it is proved for the ring of integers of a quadratic field, where every nonzero coset of the relevant lattice is anisotropic and the conclusion is immediate. At an Eichler order of level $N > 1$ the lattice $L_-$ is $N$-scaled at the level prime and has nonzero *isotropic* cosets, and there Lemma 2 computes the local factor to be a nonzero constant. So what we are contradicting is not a verifiable proof but a lemma whose only published analogue is proved under a hypothesis these lattices violate; Yang's [4, (11)] inherits the conclusion. Two later papers are worth putting next to it. Bruinier–Yang [6, (2.23)] repeat "$\kappa(m, \mu) = 0$ for $m < 0$ or $m = 0$, $\mu \neq 0$", citing Lemma 2.21, for the lattice $(\mathfrak{a}, -N/N(\mathfrak{a}))$ of an ideal of a field of odd fundamental discriminant, where $L'/L \cong \mathbb{Z}/D\mathbb{Z}$ has no nonzero isotropic element and the statement is true. Bruinier–Ehlen–Yang [7, after (3.14)], for an arbitrary binary lattice, say the opposite in general: "the Fourier coefficients of $E'_L(\tau, 0; 1)$ are computed in [KY10] ... In particular, up to a common rational scaling factor, the $\kappa(m, \mu)$ with $(m, \mu) \neq (0, 0)$ are given by logarithms of positive rational numbers" — $\kappa(0, \mu)$ for $\mu \neq 0$ included, and computed from the same local Whittaker functions [3, Thms. 4.3, 4.4] that Lemma 2 evaluates on the Eichler-order plane. So the only outside computation of these constant terms is the one we use. Everything else in the chain — the constant-term factorisation [3, Thm. 2.4], the vanishing of the incoherent series at $s = 0$, and the local structure of $L_-$ [4, Lemma 18] — is used unchanged. One point of bookkeeping decides the shape of the final statement: the quantity set to zero enters the Stokes argument [1, (33)] additively through $b_\mu(0, t)$ — the $(0, 2)$-theorem is proved as $\int \Phi(z_0, h; F)\, dh = \lim_t \big[2 \sum_\mu \sum_{m \geq 0} c_\mu(-m) b_\mu(m, t) - 2c_0(0) \log t\big]$, whose "$m = 0$ part" $2 \sum_{\mu \neq 0} c_\mu(0) b_\mu(0, t) + 2c_0(0)(b_0(0, t) - \log t)$ Lemma 2.21 then reduces to $2c_0(0)k_0(0)$, so the $m = 0$, $\mu \neq 0$ terms carry the same factor 2 as every $m > 0$ term, and in [1, Thm. 3.4(ii), Cor. 3.5(i)] the sum is over $m \geq 0$ with the single prefactor $-2/\mathrm{vol}(K_T)$; with $\mathrm{vol}(K_T) = 2/h_k$ and $\deg Z(U)_K = 2h_k$ this is $-\deg Z(U)_K/2$ on $\log\|\Psi\|^2$, i.e. Theorem B's $-|\mathrm{CM}(d)|/4$ on $\log|\psi|$, for the $m = 0$ terms as for the others — which so replacing it by $\lim_t b_\mu(0, t)$ is legitimate; but it must be replaced *wherever* Yang's (12) invokes it, and (12) calls $\kappa_\nu^-(0)$ not only at $m = 0$ but at every $m > 0$ through the vectors $x$ on the CM line with $Q(x) = m$. Proposition 6 keeps all of them.

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Throughout, $N$ is prime, $d < 0$ is a fundamental discriminant with $N \nmid d$ (so $N$ splits in $k = \mathbb{Q}(\sqrt{d})$ by Eichler's condition, the hypothesis $\mathrm{CM}(d) \neq \varnothing$ of [5, Thm. B]), $\lambda = \varphi(\sqrt{d})$, $\langle \alpha, \beta \rangle = \mathrm{tr}(\alpha\beta')$ so that $\langle \lambda, \lambda \rangle = 2|d|$, $Q = \langle \cdot, \cdot \rangle/2$, $\varphi_\nu = \mathrm{char}(\nu + \widehat{L_-})$, and $A_\nu(s, 0, v) = b_\nu(0, v)\, s + O(s^2)$.

**Lemma 1** (the lattice at the level prime)**.** *$L_{-,N}$ is the $N$-scaled hyperbolic plane, $N \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ up to a unit; $L_{-,N}^\vee / L_{-,N} \cong (\mathbb{Z}/N)^2$ with $Q(x, y) = xy/N \bmod \mathbb{Z}$, and the nonzero isotropic cosets are the $2(N-1)$ nonzero vectors on the two lines $x = 0$, $y = 0$. Moreover $L_N = L_{+,N} \oplus L_{-,N}$.*

*Proof.* [4, Lemma 18(1)]: for odd $N \nmid d$ the Gram matrix of $L_{-,N}$ is $N\,\mathrm{diag}(\epsilon_1, \epsilon_2)$ with $\epsilon_1\epsilon_2 = -d$, and $\chi_d(N) = 1$ makes $-\epsilon_1\epsilon_2$ a square, so the plane is split. For $N = 2$ (then $d \equiv 1 \bmod 8$) [4, Lemma 18(3)] gives $2\epsilon \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ directly (Yang writes out the 2-adic proof only for $2 \nmid DN$; we checked this case: $L_{-,2}$ has basis $\left(\begin{smallmatrix} -c_3/c_1 & 1 \\ 0 & c_3/c_1 \end{smallmatrix}\right), \left(\begin{smallmatrix} -c_2/c_1 & 0 \\ 2 & c_2/c_1 \end{smallmatrix}\right)$ in the notation below, with Gram matrix twice an even form with odd off-diagonal entry and determinant $\equiv -1 \bmod 8$).

For the direct sum it suffices that $\langle \lambda, L_N \rangle \subset \langle \lambda, \lambda \rangle \mathbb{Z}_N$: then the orthogonal projection $x \mapsto \frac{\langle x, \lambda \rangle}{\langle \lambda, \lambda \rangle} \lambda$ maps $L_N$ into $\mathbb{Z}_N \lambda = L_{+,N}$ and $x - \mathrm{pr}(x) \in L_N \cap \lambda^\perp = L_{-,N}$. For odd $N$, $2|d|$ is an $N$-unit. For $N = 2$: $\mathcal{O} \otimes \mathbb{Z}_2 = \left(\begin{smallmatrix} \mathbb{Z}_2 & \mathbb{Z}_2 \\ 2\mathbb{Z}_2 & \mathbb{Z}_2 \end{smallmatrix}\right)$ (remark in the proof of [4, Lemma 18]), $\lambda = \left(\begin{smallmatrix} c_1 & c_2 \\ 2c_3 & -c_1 \end{smallmatrix}\right)$, and optimality of the embedding of $\mathcal{O}_d$, $d \equiv 1 \bmod 8$, means $(1 + \lambda)/2 \in \mathcal{O} \otimes \mathbb{Z}_2$, i.e. $c_1$ odd and $c_2, c_3$ even; for $\alpha = \left(\begin{smallmatrix} a & b \\ 2c & -a \end{smallmatrix}\right) \in L_2$, $\langle \alpha, \lambda \rangle = -\mathrm{tr}(\alpha\lambda) = -2(ac_1 + bc_3 + cc_2) \in 2\mathbb{Z}_2 = 2|d|\mathbb{Z}_2$. □

Note that $L_{+,N} = \mathbb{Z}_N \lambda$ is unimodular for odd $N$ but not for $N = 2$, where $L_{+,2}^\vee / L_{+,2} \cong \mathbb{Z}/2$ is generated by $\lambda/2$ with $Q(\lambda/2) = |d|/4 \notin \mathbb{Z}_2$.

**Lemma 2** (the $m = 0$ local factor at $N$)**.** *With $X = N^{-s}$ and $\gamma_N$ the local splitting index of* [3, (4.3)]*, for $\nu \in L_{-,N}^\vee / L_{-,N}$,*

$$W_{0,N}(s, \varphi_0) = \frac{\gamma_N}{N} \cdot \frac{1 + (N-2)X}{1 - X}, \qquad W_{0,N}(s, \varphi_\nu) = \frac{\gamma_N}{N} \ (\nu \neq 0 \text{ isotropic}), \qquad W_{0,N}(s, \varphi_\nu) = 0 \ (\nu \neq 0 \text{ anisotro}$$

*Proof.* The first is [4, (15)] with $\chi_d(N) = 1$ for odd $N$, where $\frac{1 - X/N}{N - X} = \frac{1}{N}$, and [4, (16)] for $N = 2$. For the others apply [3, (4.4) and Thm. 4.3] ($N$ odd; KY's matrix $S$ is the matrix of the bilinear form with $Q = S/2$, as is Yang's Gram matrix, so $S = N\,\mathrm{diag}(\epsilon_1, \epsilon_2)$, $l_1 = l_2 = 1$, $s_0 = n/2 - 1 = 0$) to $\nu = (a/N, b/N)$ with $m = 0$: $Q(\nu) = (\epsilon_1 a^2 + \epsilon_2 b^2)/(2N)$ lies in $\mathbb{Z}_N$ exactly when $\nu$ is isotropic (for odd $N$ the 2 is a unit and may be dropped), so $W = 0$ otherwise by Thm. 4.3(i); when it is, $-\epsilon_1\epsilon_2$ being a square forces $a, b$ both units, so $H_\nu = \varnothing$, $K_0(\nu) = \min(l_i + \mathrm{ord}_N \nu_i) = 0$, and $a_\nu = \mathrm{ord}_N(t_\nu) \geq 0 = K_0$ puts us in the second case of Thm. 4.3(ii), whose sum over $0 < k \leq K_0$ is empty: $W_N(s, 0, \nu) = 1$. For $N = 2$, [3, Thm. 4.4] with the Gram matrix $2\epsilon \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, i.e. KY's (4.9) with $H = N = 0$, $M = 1$, $m_1 = 1$: $\nu' = (a/2, b/2) \notin \mathbb{Z}_2^2$ gives $M_\nu = \varnothing$, $K_0 = m_1 + \mathrm{ord}_2 \nu' = 0$, both sums in (ii) are empty, and $W_2(s, 0, \nu) = 1$ when $Q(\nu) = ab/2 \in \mathbb{Z}_2$, i.e. at the two nonzero isotropic cosets. In every case the normalising constant of [3, (4.4)] is $\gamma_N |\det S|_N^{1/2} = \gamma_N / N$, the same for all $\nu$, so it cancels

in the ratio used below. Lemma lem:m0loc of the preprint is the same object in a counting normalisation (its unimodular calibration is Thm. 4.3 with $l_i = 0$), and its third and fourth rows are these values. □

**Lemma 3** (no other nonzero coset contributes)**.** *For $p \neq N$ every nonzero $\nu_p \in L_{-,p}^\vee/L_{-,p}$ is anisotropic, $Q(\nu_p) \notin \mathbb{Z}_p$. Consequently $W_{0,p}(s, \varphi_{\nu,p}) = 0$ for such $\nu_p$, and a nonzero isotropic coset of $L_-^\vee/L_-$ is supported at $N$.*

*Proof.* $W_{0,p} = 0$ follows from anisotropy by [3, Thm. 4.3(i)] ($p$ odd) and [3, Thm. 4.4(i)] ($p = 2$). Anisotropy is read off [4, Lemma 18]; $d$ fundamental gives $p \parallel d$ for odd $p \mid d$.

$p$ odd, $p \nmid dDN$: unimodular. $p$ odd, $p \mid d$: Gram $\operatorname{diag}(\epsilon_1, \epsilon_2)$ with, say, $\epsilon_2 = p\epsilon_2'$; $L^\vee/L \cong \mathbb{Z}/p$ generated by $(0, 1/p)$, and $Q(0, b/p) = \epsilon_2' b^2/(2p) \notin \mathbb{Z}_p$ for $p \nmid b$. $p$ odd, $p \mid D$, $p \nmid d$: Gram $p\operatorname{diag}(\epsilon_1, \epsilon_2)$ with $\epsilon_1\epsilon_2 = -d$; $p$ is inert in $k$, so $d$ is a non-square mod $p$ (the Hilbert-symbol condition $(-\mu_1, -\mu_2 p)_p = -1$ in Yang's proof), and $Q(a/p, b/p) = (\epsilon_1 a^2 + \epsilon_2 b^2)/(2p) \in \mathbb{Z}_p$ forces $a \equiv b \equiv 0$.

$p = 2 \nmid d$. If $d \equiv 1 \bmod 8$ then $2 \nmid D$ and, as $2 \neq N$, Lemma 18(3) gives $\epsilon \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, unimodular. If $d \equiv 5 \bmod 8$, Lemma 18(4) gives $\epsilon \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ for $2 \nmid D$, unimodular, and $2\epsilon \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ for $2 \mid D$, where $L^\vee/L \cong (\mathbb{Z}/2)^2$ and $Q(a/2, b/2) = \epsilon(a^2 + ab + b^2)/2$ with odd numerator for $(a, b) \not\equiv (0, 0)$. (This case is again one Yang only sketches; with $(1+\lambda)/2$ in the maximal order, $a, b, c$ are all odd in $\lambda = aI + bJ + cIJ$ below, and the elimination below gives diagonal entries $\equiv 2 \bmod 8$, a unit off-diagonal and determinant $\equiv 3 \bmod 8$.)

$p = 2 \mid d$: Lemma 18(2) gives $2\operatorname{diag}(\epsilon_1, \epsilon_2)$ with $\epsilon_1\epsilon_2 = -d/4$, $Q(x, y) = \epsilon_1 x^2 + \epsilon_2 y^2$, and $d$ fundamental means $d/4 \equiv 2, 3 \bmod 4$. If $d \equiv 12 \bmod 16$ both $\epsilon_i$ are units with $\epsilon_1 \equiv \epsilon_2 \bmod 4$, and the three nonzero cosets of $(\mathbb{Z}/2)^2$ have $Q = \epsilon_1/4$, $\epsilon_2/4$, $(\epsilon_1+\epsilon_2)/4 \equiv \epsilon_1/2$. If $d \equiv 8 \bmod 16$, say $\epsilon_2 = 2\epsilon'$; $L^\vee/L \cong \mathbb{Z}/2 \times \mathbb{Z}/4$ and $Q(a/2, b/4) = \epsilon_1 a^2/4 + \epsilon' b^2/8$ has 2-adic valuation $-3$ for $b$ odd, $-2$ or $-1$ for $b \equiv 2 \bmod 4$, and equals $\epsilon_1/4$ for $b \equiv 0$, $a$ odd. None is in $\mathbb{Z}_2$.

Yang writes out the proof of Lemma 18(2) only for $2 \nmid DN$; the case $2 \mid N$, $2 \mid d$ does not occur here, and the case $2 \mid D$, $2 \mid d$ is proved as follows. $B \otimes \mathbb{Q}_2 = \left(\frac{-1,-1}{\mathbb{Q}_2}\right)$ with maximal order $\mathbb{Z}_2 + \mathbb{Z}_2 I + \mathbb{Z}_2 J + \mathbb{Z}_2(1 + I + J + IJ)/2$, whose trace-zero part is $L_2 = \mathbb{Z}_2 I + \mathbb{Z}_2 J + \mathbb{Z}_2 IJ$ with $Q = x^2 + y^2 + z^2$. Since $d \equiv 0 \bmod 4$, $\mathcal{O}_d = \mathbb{Z}[\sqrt{d}/2]$ and $L_+ = \mathbb{Z}\lambda_0$ with $\lambda_0 = \lambda/2 = aI + bJ + cIJ$, $a^2 + b^2 + c^2 = m := -d/4 \equiv 1$ or $2 \bmod 4$, and $L_{-,2} = \{xI + yJ + zIJ : ax + by + cz = 0\}$. If $m \equiv 1 \bmod 4$ exactly one of $a, b, c$ is odd, say $a$; eliminating $x = -(by + cz)/a$, $Q|_{L_{-,2}} = u_1 y^2 + 2w\, yz + u_2 z^2$ with $u_1 = b^2/a^2 + 1$, $u_2 = c^2/a^2 + 1$ units and $w = bc/a^2 \in 4\mathbb{Z}_2$; $y' = y + (w/u_1)z$ is an integral change of basis and $Q = u_1 y'^2 + (u_2 - w^2/u_1)z^2$ with $u_1 u_2 - w^2 = m/a^2$. If $m \equiv 2 \bmod 4$ two of $a, b, c$ are odd, say $a, b$; then $u_1 \in 2\mathbb{Z}_2^\times$, $u_2$ is a unit, $w \in 2\mathbb{Z}_2$, and completing the square on $z$ instead gives $Q = u_2 z'^2 + (u_1 - w^2/u_2)y^2$, product again $m/a^2$. Rescaling one basis vector by the unit $a$ makes the product $-d/4$ exactly, which is Lemma 18(2). □

**Proposition 4** (the dropped coefficient)**.** *Let $N$ be prime, $N \nmid d$, and $\nu \in L_-^\vee/L_-$ a nonzero isotropic coset. Then $b_\nu(0, v) = -\log N/(N-1)$ for every $v$, hence*

$$\kappa_\nu^-(0) := \lim_{v \to \infty} b_\nu(0, v) = -\frac{\log N}{N-1}.$$

*Proof.* By [3, Thm. 2.4], as used in [1, proof of Lemma 2.21] and [4, (14)], the constant term is $E_0(\tau, s; \varphi_\nu, +1) = v^{s/2}\varphi_\nu(0) + W_{0,\infty}(\tau, s)\prod_p W_{0,p}(s, \varphi_{\nu,p})$ with $W_{0,\infty}$ finite at $s = 0$ ([3, Prop. 2.3(iii)] with $l = 1$; only finiteness is used below). Put $\Phi(s) = E_0(\tau, s; \varphi_0, +1) - v^{s/2}$.

For $\nu \neq 0$, $\varphi_\nu(0) = 0$, and $\varphi_{\nu,p} = \varphi_{0,p}$ for $p \neq N$ (Lemma 3), so for $\operatorname{Re} s$ large, hence by meromorphic continuation,

$$E_0(\tau, s; \varphi_\nu, +1) = \Phi(s)\,\frac{W_{0,N}(s, \varphi_\nu)}{W_{0,N}(s, \varphi_0)} = \Phi(s)\,\frac{1 - N^{-s}}{1 + (N-2)N^{-s}}$$

by Lemma 2. (The Euler product does not converge at $s = 0$; $\Phi$ and the identity are defined through Thm. 2.4 and continued. The pole of $W_{0,N}(s, \varphi_0)$ at $s = 0$ is cancelled by the missing Euler factor $1 - N^{-s}$ of $L^S(s, \chi_d)$ there, and $W_{0,N}(s, \varphi_0)$ has no zero near $s = 0$.) The series $E(\tau, s; \varphi_0, +1)$ is holomorphic at $s = 0$ ([3, Cor. 2.5]) and incoherent, so it vanishes there ([4], the paragraph before (11)); its constant term gives $\Phi(0) = -1$. The last factor vanishes at $s = 0$ — as it must, $E(\tau, s; \varphi_\nu, +1)$ being incoherent — with derivative $\log N/(N-1)$, so $A_\nu(s, 0, v) = -\frac{\log N}{N-1}s + O(s^2)$, with no $v$ remaining: the $v$-dependence of $\Phi$ enters at order $s$, multiplying a term already $O(s)$.

As a cross-check, [4, (14)] writes $\Phi(s) = -v^{-s/2}\frac{\Lambda(s,\chi_d)}{\Lambda(s+1,\chi_d)}G(s)$ with $G = \prod_{p|r} G_p$ over the non-unimodular primes; the functional equation of the completed $L$-function of the odd primitive character $\chi_d$ gives $\Lambda(0) = \Lambda(1)$, and $G(0) = 1$ is read off (15)–(17) factor by factor, so $\Phi(0) = -1$ again. The sign agrees with [4, Lemma 20], where $-\frac{d}{ds}\log G_N\big|_{s=0} = \log N$ at the zero coset. □

**Lemma 5** (isotropic cosets of $L^\vee/L$)**.** *For $N$ prime the isotropic cosets of $L^\vee/L$ are supported at $N$ and number $2N-1$: under $L_N = L_{+,N} \oplus L_{-,N}$ they are the cosets $0 \oplus \nu$ with $\nu$ an isotropic coset of $L_{-,N}^\vee/L_{-,N}$. All $2N-2$ nonzero ones carry the same constant term $c_\eta(0)$ of $F_f$.*

*Proof.* A coset is isotropic iff each $p$-part is, and $L^\vee/L$ does not depend on $d$, so the computation is on $L_p$ itself. For $p$ odd, $p \nmid DN$, $L_p$ is unimodular. For $p$ odd, $p \mid DN$, the proof of [4, Lemma 18] gives a basis of $L_p$ with Gram matrix $\operatorname{diag}(2\mu_1, 2\mu_2 p, 2\mu_1\mu_2 p)$, $\mu_i$ units, so $(L^\vee/L)_p \cong (\mathbb{Z}/p)^2$ with $Q(ae_2/p + be_3/p) = \mu_2(a^2 + \mu_1 b^2)/p$: anisotropic iff $-\mu_1$ is a non-square mod $p$, i.e. iff $(-\mu_1, -\mu_2 p)_p = -1$, which Yang's proof identifies with $p \mid D$; for $p \mid N$ it is the hyperbolic plane over $\mathbb{F}_p$, consistent with Lemma 1. For $p = 2 \nmid DN$, $L_2$ is the trace-zero part of $M_2(\mathbb{Z}_2)$, with Gram $(-2) \oplus \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ in the basis $\operatorname{diag}(1, -1), e_{12}, e_{21}$, so $(L^\vee/L)_2 \cong \mathbb{Z}/2$ generated by $\frac{1}{2}\operatorname{diag}(1, -1)$ with $Q = -1/4$. For $p = 2 \mid D$, $L_2 = \mathbb{Z}_2 I + \mathbb{Z}_2 J + \mathbb{Z}_2 IJ$ as in the proof of Lemma 3, $(L^\vee/L)_2 \cong (\mathbb{Z}/2)^3$ with $Q = (a^2 + b^2 + c^2)/4 \in \{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ on nonzero elements. At $p = N$ odd, $L_{+,N}$ is unimodular and $(L^\vee/L)_N = L_{-,N}^\vee/L_{-,N}$, with $2N-1$ isotropic vectors. At $N = 2$, the orthogonal direct sum gives $(L^\vee/L)_2 = L_{+,2}^\vee/L_{+,2} \oplus L_{-,2}^\vee/L_{-,2} \cong \mathbb{Z}/2 \oplus (\mathbb{Z}/2)^2$; the elements $\lambda/2 \oplus \nu$ have $Q = |d|/4 + Q(\nu)$ with $Q(\nu) \in \{0, \frac{1}{2}\}$ mod $\mathbb{Z}$ and $|d|$ odd, never integral, so the isotropic elements are the $0 \oplus \nu$ with $\nu$ isotropic: again $2N - 1 = 3$.

For the constant terms: an isometry $\sigma$ of the discriminant form permutes the $e_\eta$, fixes $e_0$, and commutes with $\rho_L(S)$ and $\rho_L(T)$ because it preserves $Q$ and the pairing, hence with every $\rho_L(\tilde{\gamma})$; so $\sigma F_f = F_f$ and $c_{\sigma\eta}(0) = c_\eta(0)$. The orthogonal group of the hyperbolic plane over $\mathbb{F}_N$ contains the torus $(x, y) \mapsto (ax, a^{-1}y)$ and the swap, and acts transitively on the $2(N-1)$ nonzero isotropic vectors; these isometries of $(L^\vee/L)_N$ extend by the identity on the other $p$-parts. □

**Proposition 6** (the restored terms of Theorem B)**.** *Let $N$ be prime, $N \nmid d$, $d$ fundamental, $\tau_d$ off the divisor of $\Psi_F$ (Theorem B's own hypothesis; it is used for the $x \neq 0$ terms, see Remark 9, and Remark 25 treats a point on the divisor), and let $\nu$ run over the $2N-2$ nonzero isotropic cosets of $L_-^\vee/L_-$, identified with those of $L^\vee/L$ by Lemma 5. Then $\kappa_\nu(0) = \kappa_\nu^-(0) =$*

$-\log N/(N-1)$, *and the terms of* [5, Thm. B] *that* [4, (11)] *sets to zero are, in total,*

$$-\frac{|\mathrm{CM}(d)|}{4}\sum_{\nu}\kappa_{\nu}^{-}(0)\sum_{x\in L^{\vee}\cap\mathbb{Q}\lambda}c_{[x+\nu]}(-Q(x)) \;=\; \frac{|\mathrm{CM}(d)|}{4}\,\frac{\log N}{N-1}\sum_{\nu}\sum_{x\in L^{\vee}\cap\mathbb{Q}\lambda}c_{[x+\nu]}(-Q(x)),$$

*with* $[\,\cdot\,]$ *the class in* $L^{\vee}/L$. *The* $x=0$ *term of each inner sum is* $c_{[\nu]}(0)$, *and those* $2N-2$ *values agree (Lemma 5), so the* $x=0$ *part of the total is* $\frac{1}{2}|\mathrm{CM}(d)|c_{\eta}(0)\log N$; *the* $x\neq 0$ *terms are identified in Lemma 7(3) and Remark 9, and are not assumed here to be independent of* $\nu$. *In particular, if*

(H) $c_{[x+\nu]}(-Q(x))=0$ *for every* $x\neq 0$ *in* $L^{\vee}\cap\mathbb{Q}\lambda$ *and every nonzero isotropic* $\nu$,

*the correction is* $|\mathrm{CM}(d)|\cdot\frac{1}{2}c_{\eta}(0)\log N$, *i.e.* $\mathrm{mult}(f)\sum_{p\mid N/(N,d)}\log p$ *per CM point with* $\mathrm{mult}(f)=\frac{1}{2}c_{\eta}(0)$, *which is* Observation obs:mult of the preprint. *(Theorem B constrains the sum over* $\mathrm{CM}(d)$; *the correction does not depend on the point, so dividing by* $|\mathrm{CM}(d)|$ *is the convention of* Section sec:chain of the preprint.*)*

*Proof.* Theorem B sums $c_{\eta}(-m)\kappa_{\eta}(m)$ over $\eta\in L^{\vee}/L$ and $m\geq 0$, with $\kappa_{\eta}(m)$ as in Section sec:chain of the preprint: $\kappa_{\eta}(m)=\sum_{\mu\in L/(L_{+}+L_{-})}\sum_{x\in\eta_{+}+\mu_{+}+L_{+}}\kappa^{-}_{\eta_{-}+\mu_{-}}(m-Q(x))$, where $\eta=\eta_{+}+\eta_{-}$ is the decomposition in $L_{+}^{\vee}\oplus L_{-}^{\vee}\supset L^{\vee}$. The pairs $(\eta,\mu)$ run over $L^{\vee}/(L_{+}\oplus L_{-})$ exactly once, and every factor depends only on $y=\eta+\mu$; the terms in which $\kappa^{-}$ is evaluated at 0 are those with $m=Q(x)$ ($L_{+}$ is positive definite and $\kappa_{\nu}^{-}(m)=0$ for $m<0$, so no other term involves $\kappa_{\nu}^{-}(0)$), and their total is

$$\sum_{y\in L^{\vee}/(L_{+}\oplus L_{-})}\;\sum_{x\in y_{+}+L_{+}}c_{[y]}(-Q(x))\,\kappa^{-}_{[y_{-}]}(0) \;=\; \sum_{\nu\in L_{-}^{\vee}/L_{-}}\kappa_{\nu}^{-}(0)\sum_{\substack{x\in L_{+}^{\vee}\\ x+\nu\in L^{\vee}}}c_{[x+\nu]}(-Q(x)),$$

the inner sum over all $x$ in the coset, both signs included. By Lemma 3 and Proposition 4, among nonzero $\nu$ the coefficient $\kappa_{\nu}^{-}(0)$ is nonzero only for $\nu$ isotropic, supported at $N$, where it equals $-\log N/(N-1)$ (the $\nu=0$ terms are the standard ones, unchanged). Such a $\nu$ lies in $L^{\vee}$: at $N$ by $L_{N}^{\vee}=L_{+,N}^{\vee}\oplus L_{-,N}^{\vee}$, which is Lemma 1, and elsewhere because $\nu_{p}\in L_{-,p}\subset L_{p}$. So the condition $x+\nu\in L^{\vee}$ reads $x\in L^{\vee}\cap\mathbb{Q}\lambda$ and does not depend on $\nu$. The $x=0$ term is $c_{[\nu]}(0)$, with $[\nu]$ a nonzero isotropic coset of $L^{\vee}/L$; the $2N-2$ values agree by Lemma 5. This gives the display, with the prefactor $-|\mathrm{CM}(d)|/4$ of Theorem B and $-\frac{1}{4}(2N-2)(-\log N/(N-1))=\frac{1}{2}\log N$; the restored terms are *added*, as Observation obs:mult of the preprint requires.

For $\kappa_{\nu}(0)$ itself take $\eta=\nu$ and $m=0$: only $x=0$ contributes, which needs $\mu_{+}\in L_{+}$ since $\nu_{+}=0$; then $\mu_{-}=\mu-\mu_{+}\in L\cap U=L_{-}$, so $\mu=0$ in $L/(L_{+}+L_{-})$ is the only class, and for it $\eta_{-}+\mu_{-}\equiv\nu$ modulo $L_{-}$. Hence $\kappa_{\nu}(0)=\kappa_{\nu}^{-}(0)$. (For $\eta=0$ the same count gives $\mu=0$ alone, which is [4, Lemma 20]'s $\kappa_{0}(0)=\kappa_{0}^{-}(0)$, unchanged.) □

The same accounting is visible in Schofer's own reduction: the $(n,2)$ theorem is proved by applying the $(0,2)$ theorem to $\langle F,\theta_{+}\rangle$, whose constant term at the coset $\eta_{-}+\mu_{-}$ is $\sum_{x\in\eta_{+}+\mu_{+}+L_{+}}c_{\eta}(-Q(x))$ [1, Cor. 3.2], and the $m=0$ part of the $(0,2)$ proof is $2\sum_{\mu\neq 0}c_{\mu}(0)b_{\mu}(0,t)$ with *that* constant term, before Lemma 2.21 sets it to zero.

**Lemma 7** (when (H) holds)**.** *Let* $F_{f}$ *be built from a scalar form* $f$ *on* $\Gamma_{0}(M)$ *with poles only at the cusps* $\infty$ *and* 0, *as in* [5, (6)]. *By* [5, Lemma 24], *in the normalisation of* Section sec:data of the preprint *(where it is checked numerically on every form used), the principal part of* $F_{f}$ *at a nonzero coset* $\eta$ *is that of* $f$ *at the cusp* 0, $c_{\eta}(-n/M)$ *being the coefficient of* $q^{-n/M}$ *in* $f|S$, *and depends on* $\eta$ *only through* $Q(\eta)$ mod $\mathbb{Z}$; *at* $\eta=0$ *the principal part of* $f$ *at* $\infty$ *is added. Hence:*

(1) *Let* $L_+ = \mathbb{Z}\lambda_0$, *with* $\lambda_0 = \lambda$ *or* $\lambda/2$ *according as* $d \equiv 1$ *or* $0 \bmod 4$ [4, proof of Lemma 18], *so* $Q(\lambda_0) = |d|$ *or* $|d|/4$, *and* $L^\vee \cap \mathbb{Q}\lambda = \mathbb{Z}\lambda_0/g$ *with* $g$ *the content of* $\langle \lambda_0, \cdot \rangle$ *on* $L$: $g$ *divides* $2Q(\lambda_0)$, *is prime to odd* $N$ *and to the primes* $p \mid D$, $p \nmid d$, *is divisible exactly by* 2 *when* $d \equiv 1 \bmod 4$ *or* $2 \mid D$, *and exactly by* $p$ *for each odd* $p$ *dividing both* $D$ *and* $d$ *(there* $p \mid c_1$ *and* $c_2$ *is a unit in Yang's basis). If the pole of* $f$ *at the cusp* 0 *has order below* $M \cdot \min_{j \geq 1} j^2 Q(\lambda_0)/g^2 = MQ(\lambda_0)/g^2$ *in* $q^{1/M}$, *then (H) holds at* $d$. *The threshold can be small: on* $X_0^{15}(2)$ *at* $d = -15$, $g = 30$ *and it is* $1/60$ *in powers of* $q$, *i.e. a single power of the local parameter* $q^{1/60}$.

(2) *In particular (H) holds at every* $d$ *when* $f$ *is holomorphic at the cusp* 0. *A pole of order below* $M$ *(one power of* $q$*) removes only the terms with* $x \in L$; *those with* $x \notin L$ *sit at the exponents* $Mj^2Q(\lambda_0)/g^2$, $g \nmid j$, *which lie below* $M$ *as soon as* $g$ *is large.*

(3) *For* $x \notin L$ *the coefficient* $c_{[x+\nu]}(-Q(x))$ *equals* $c_{[x]}(-Q(x))$, *both cosets being nonzero (the* $N$*-part of* $x + \nu$ *is not in* $L_N$, *the sum there being direct) with the same norm; for* $x \in L$ *it is the cusp-*0 *part* $c_0(-Q(x)) - c_\infty(-Q(x))$ *of the coefficient at the zero coset.*

**Remark 8** (non-fundamental $d$ with $N \nmid f$, and the input)**.** The hypothesis "$d$ fundamental" in Proposition 6 is used only through the local lattice at $N$. If $d = f^2 d_0$ with $N \nmid f$ then $R_f \otimes \mathbb{Z}_N = \mathcal{O}_k \otimes \mathbb{Z}_N$ and $L_{-,N}$ is the plane of the fundamental case, so Lemma 1, Lemma 2 and Proposition 4 apply verbatim at $N$ with $N \nmid d_0$ in place of $N \nmid d$, and the terms at $N$ are those of the proposition; the primes of the conductor contribute separately, by Proposition 19, the contributions being additive over the primes exactly as in Proposition 10 (a coset supported at two primes has a double zero). This is the case the code evaluates when a non-fundamental $d$ is asked for at a level prime not dividing $f$. The input $F_f$ is Guo–Yang's: $f$ a weight-$\frac{1}{2}$ eta quotient on $\Gamma_0(M)$ with poles only at the cusps $\infty$ and 0 ("$\{\infty, 0\}$-weakly holomorphic"), and its principal part is [5, Lemma 24], a lemma proved there for exactly this construction (the proof uses only that $P(f|\gamma) = 0$ at the other cusps and the formula for $\rho_L(S^{-1})e_\delta$); the pipeline's pool imposes the cusp condition, so the lemma, and with it Lemma 5 and Proposition 6, apply to every form the pipeline produces.

**Remark 9** (the $x \neq 0$ terms under Theorem B's own hypothesis)**.** Theorem B is applied at $\tau_d \notin \operatorname{supp} \operatorname{div} \psi_F$. By Borcherds' theorem in the form [4, Thm. A(2)], the order of $\Psi_F$ at $\tau_d$ is $\sum_{0<r,\, r\lambda \in L^\vee} c_{r\lambda}(-r^2\langle\lambda,\lambda\rangle/2) = \sum_{x \in L^\vee \cap \mathbb{Q}\lambda,\, x>0} c_{[x]}(-Q(x))$, and $c_{[-x]} = c_{[x]}$; so $\sum_{x \neq 0} c_{[x]}(-Q(x)) = 0$ at an evaluated point, and by Lemma 7(3)

$$\sum_{x \neq 0} c_{[x+\nu]}(-Q(x)) = \sum_{x \neq 0} c_{[x]}(-Q(x)) - \sum_{x \in L \cap \mathbb{Q}\lambda,\, x \neq 0} c_\infty(-Q(x)) = -2 \sum_{k \geq 1} c_\infty\big(-k^2 Q(\lambda_0)\big),$$

with $c_\infty(-n)$ the coefficient of $q^{-n}$ in $f$ at $\infty$. So the full correction per CM point is

$$\log N \Big( \tfrac{1}{2} c_\eta(0) - \sum_{k \geq 1} c_\infty\big(-k^2 Q(\lambda_0)\big) \Big) :$$

the $x \neq 0$ terms are nonzero exactly when $f$ has a pole at $\infty$ of order $k^2 Q(\lambda_0)$ whose contribution to the order at $\tau_d$ is cancelled by the cusp-0 side. Such a point is a legitimate point of evaluation — Theorem B's hypothesis is the vanishing of the order, not of each term — with a correction different from $\frac{1}{2} c_\eta(0)$ and dependent on $d$. Under Lemma 7(1), in particular when $f$ is holomorphic at the cusp 0, no such cancellation is possible; otherwise the sum has to be evaluated at each point, which costs a few coefficient lookups, and the accompanying code now does so (`M0InfinityPoleSum`).

On the data: the input forms of every even-$D$ base are holomorphic at the cusp 0, so (H) holds there outright. On the six odd-$D$ bases with prime $N > 1$ in our model set ($X_0^{15}(2)$,

$X_0^{21}(2)$, $X_0^{35}(2)$, $X_0^{39}(2)$, $X_0^{51}(2)$, $X_0^{57}(2)$) the pole at the cusp 0 has order up to $\frac{5}{4}M$ (on $X_0^{39}(2)$), so (H) can fail; there the order at $\tau_d$ and the $x \neq 0$ sum were computed from the two cusp expansions for every form and every fundamental $d$ at which the term fires, and the $x \neq 0$ sum was zero at every point off the divisor. This is a structural fact of these forms rather than a tested cancellation — the principal part of each at the cusp 0 is a single monomial whose exponent is never of the form $Mj^2Q(\lambda_0)/g^2$ — and on $X_0^{39}(2)$ and $X_0^{51}(2)$ the forms as returned by the search have non-integral principal parts at $\infty$, outside Theorem B's hypothesis $c_\eta(-m) \in \mathbb{Z}$, so those two bases are not evidence either way. The agreement of Observation obs:mult of the preprint could therefore not have detected the $x \neq 0$ terms; it is Proposition 6 that accounts for them.

The same argument settles the two cases the hypotheses exclude.

**Proposition 10** (composite squarefree level)**.** *Let $N$ be squarefree, $d$ fundamental, and for a nonzero isotropic $\nu \in L_-^\vee/L_-$ let $S(\nu) = \{p \mid N : \nu_p \neq 0\}$; then $S(\nu)$ is nonempty and consists of primes not dividing $d$, and*

$$\kappa_\nu^-(0) = \begin{cases} -\log p/(p-1) & \text{if } S(\nu) = \{p\}, \\ 0 & \text{if } |S(\nu)| \geq 2. \end{cases}$$

*Consequently, under (H) at every prime, the restored terms of Theorem B contribute $\sum_{p|N,\, p\nmid d} \frac{1}{2} c_{\eta_p}(0) \log p$ per CM point, where $c_{\eta_p}(0)$ is the common constant term of the $2p-2$ cosets supported exactly at $\{p\}$. Of the $2^{\omega(N)} - 1$ functionals of* Section sec:determined of the preprint*, the multiplier at the prime $p$ is the one attached to the support class $\{p\}$; the classes supported at two or more primes do not enter.*

*Proof.* At $p \mid N$ with $p \nmid d$ the local plane is the $p$-scaled split plane (Lemma 1, prime by prime; Eichler's condition makes $p$ split), and at every other prime — $p \nmid N$, or $p \mid N$ with $p \mid d$ — every nonzero local coset is anisotropic (Lemma 3 and Remark 11). So $S(\nu)$ is as stated, and for $\operatorname{Re} s$ large

$$E_0(\tau, s; \varphi_\nu, +1) = \Phi(s) \prod_{p \in S(\nu)} \frac{1 - p^{-s}}{1 + (p-2)p^{-s}}$$

by Lemma 2 at each $p \in S(\nu)$, with $\Phi(0) = -1$ as in the proof of Proposition 4 (the poles of $W_{0,p}(s, \varphi_0)$ at the primes of $N$ are cancelled by the missing Euler factors of $L^S(s, \chi_d)$). The product has a zero of order $|S(\nu)|$ at $s = 0$, whose first-order coefficient is $\log p/(p-1)$ when $S(\nu) = \{p\}$ and 0 otherwise. The count $\kappa_\nu(0) = \kappa_\nu^-(0)$ is as in Proposition 6: with $\eta = \nu$ and $m = 0$, $\mu_+ \in L_+$ forces $\mu = 0$. The cosets with support exactly $\{p\}$ are the nonzero isotropic vectors of the plane at $p$ with zero components elsewhere, one orbit of its orthogonal group extended by the identity, so they share $c_{\eta_p}(0)$ (Lemma 5); there are $2p-2$ of them, and $-\frac{1}{4}(2p-2)(-\log p/(p-1)) = \frac{1}{2}\log p$. The $x \neq 0$ terms are handled prime by prime exactly as in Lemma 7 and Remark 9. □

**Remark 11** (a prime of $N$ dividing $d$)**.** If $p \mid N$ and $p \mid d$, nothing is dropped at $p$: $L_{-,p}$ is anisotropic, so no nonzero isotropic coset is supported there and $W_{0,p}(s, \varphi_{\nu,p}) = 0$ for every $\nu_p \neq 0$. For odd $p$ this is [4, Lemma 18(1)] with $p \mid (DN, d)$, Gram $\operatorname{diag}(\epsilon_1, \epsilon_2)$ with $p \parallel \epsilon_2$, as in the proof of Lemma 3. For $p = 2$, a case Yang omits, write $\lambda_0 = \lambda/2 = \begin{pmatrix} c_1 & c_2 \\ 2c_3 & -c_1 \end{pmatrix}$ in $\mathcal{O} \otimes \mathbb{Z}_2 = \begin{pmatrix} \mathbb{Z}_2 & \mathbb{Z}_2 \\ 2\mathbb{Z}_2 & \mathbb{Z}_2 \end{pmatrix}$, $c_1^2 + 2c_2c_3 = d/4 \equiv 2, 3 \bmod 4$, and $L_{-,2} = \{\begin{pmatrix} a & b \\ 2c & -a \end{pmatrix} : ac_1 + bc_3 + cc_2 = 0\}$ with $Q = -(a^2 + 2bc)$ (the reduced norm; only its integrality is used below). If $d/4 \equiv 3 \bmod 4$ then

$c_1, c_2, c_3$ are odd; eliminating $a$ gives a form $2(uy^2 + 2v\,yz + u'z^2)$ with $u, u'$ units and $v$ even, hence $2\,\mathrm{diag}(\epsilon_1, \epsilon_2)$ with $\epsilon_1\epsilon_2 = -(d/4)/c_1^2 \equiv 1 \bmod 4$, and the three nonzero cosets of $(\mathbb{Z}/2)^2$ have $Q \equiv \epsilon_1/4, \epsilon_2/4, \epsilon_1/2$, none integral. If $d/4 \equiv 2 \bmod 4$ then $c_1$ is even and $c_2, c_3$ odd; eliminating $b$ gives $2(a^2 - 2(c_1/c_3)ac - 2(c_2/c_3)c^2)$, hence $2\,\mathrm{diag}(1, 2\epsilon')$ with $\epsilon'$ a unit, whose discriminant group $\mathbb{Z}/2 \times \mathbb{Z}/4$ has $Q(a/2, b/4) = a^2/4 + \epsilon' b^2/8$, never integral on a nonzero element. This is the firing rule $\sum_{p|N/(N,d)} \log p$: the primes of $N$ dividing $d$ contribute nothing, for the reason the recipe assumed at all of them.

For $m > 0$ the same prime enters through the local Whittaker polynomial of the anisotropic plane, which the routine computes from [3, Thms. 4.3, 4.4] on the actual $L_{-,p}$; those theorems need no unimodularity, unlike Schofer's closed form [1, Thm. 4.1], which the routine does not use. Forcing every published point at which a level prime divides $d_0$ on the sixteen Guo–Yang tables with $N > 1$ — 104 points — through the table comparison gives agreement at all of them (`vvdata/weyl-campaign/noncoprime/noncop.m` on the campaign branch; the one disagreement, $X_0^{21}(2)$ at $-420$, is the misprint of Remark 25). So the values at such points need no separate treatment; the pipeline's coprimality filter keeps them out of the Borcherds search only because admitting them changes its anchors and hence the forms.

**Remark 12** (the composite level in the code)**.** Proposition 10 is what the routine now evaluates at a squarefree level: the multiplier at the prime $p$ is $\frac{1}{2}c_{\eta_p}(0)$ with $\eta_p$ any nonzero isotropic coset of order $p$ in $L^\vee/L$ — the support of a coset being the set of primes dividing its order — and the restored term per CM point is $\sum_{p|N} \log p \left(\frac{1}{2}c_{\eta_p}(0) - \sum_{x \neq 0}\right)$, the pole sum of Remark 9 being the same at every prime (the routine `MOMultipliersBySupport` returns the multipliers indexed by the primes of $N$, checks that the constant terms agree on each support class, and refuses a level that is not squarefree; at a prime level it is the old routine). What was checked before any composite-level base existed, and is the direct test of the coset sums: at the six bases with a genus-zero star curve and $DN = 210$ — $X_0^6(35)$, $X_0^{10}(21)$, $X_0^{14}(15)$, $X_0^{15}(14)$, $X_0^{21}(10)$, $X_0^{35}(6)$, all with $|L^\vee/L| = 88200$ and $M = 420$ — the nonzero isotropic cosets number exactly $2p - 2$ on each class $\{p\}$ and $(2p-2)(2q-2)$ on $\{p, q\}$, as the proof says; and on $X_0^6(35)$, for four weight-$\frac{1}{2}$ eta quotients of level 420 with the character of the Borcherds input space but poles at every cusp — so not inputs of Theorem B, whose $F_f$ needs $f$ $\{\infty, 0\}$-weakly holomorphic for [5, Lemma 24] and the pole sum of Remark 9; what is tested is the coset sum $c_\eta(0) = \sum_w \rho(w^{-1})e_0[\eta]\, a_0(f|w)$ alone, not a Schofer value — the constant terms are constant on each of the three classes $\{5\}$, $\{7\}$, $\{5, 7\}$, rational, and different from class to class — $-\frac{23}{4}$, $-\frac{39}{4}$, 0 on one of them, 78, 64, 0 on another — so a single multiplier, the prime-level rule, would be wrong here (`composite/compmult5.m` on the campaign branch; about 45 minutes, almost all of it the Weil representation on 88200 cosets). The pipeline's own inputs satisfy the cusp condition by construction. The same routine refuses an eta quotient outside that space, as it should: the coset sum is then not well defined, and the two-point and class-constancy checks fail at the first word where the constant term is nonzero (`compmult.m`–`compmult4.m`, weight 0 or the wrong character). The three functionals of Section sec:determined of the preprint are therefore computed, and the one attached to $\{p\}$ is used at $p$. Three composite-level bases have since been built with it on this branch, $X_0^6(35)$, $X_0^{10}(21)$ and $X_0^{14}(15)$: every model of genus at least one has the point counts of the trace formula over $\mathbf{F}_p$ and $\mathbf{F}_{p^2}$ at six good primes, and every genus-one Jacobian is semistable with conductor divisible by $D$ and dividing 210, as Jacquet–Langlands predicts. No published CM value exists at a composite level, so this is the only outside evidence for the values there.

**Remark 13** (computing $c_\eta(0)$ exactly)**.** The constant term $c_\eta(0) = \sum_w \rho(w^{-1})e_0[\eta]\, a_0(f|w)$, a sum over the cosets $w$ of $\Gamma_0(M)\backslash SL_2(\mathbf{Z})$, is an algebraic number, and it can be computed with

no transcendental step. The per-coset product is well defined on $\Gamma_0(M)\backslash SL_2(\mathbf{Z})$ (both factors change by the character of $f$ on $\Gamma_0(M)$ under left multiplication, and $e_0$ is an eigenvector for that character), and it is constant on each class $g = \gcd(c, M)$ of the bottom row of $w$ for two reasons. Right multiplication by $T$ leaves $a_0(f|w)$ unchanged and multiplies the $\eta$-component of $\rho(w^{-1})e_0$ by $\mathbf{e}(\pm Q(\eta)) = 1$, $Q(\eta)$ being an integer at the isotropic $\eta$ in question, so the product is a function of the cusp $\Gamma_0(M)w\infty$ alone; and the cusps of $\Gamma_0(M)$ with $\gcd(c, M) = g$ number $\varphi(\gcd(g, M/g))$, which is 1 for every $g$ because $M$ is four times an odd squarefree number, so each class is a single cusp. One representative $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $c > 0$ per class therefore suffices, both factors taken for the lift $(\gamma, \sqrt{c\tau + d})$ of the metaplectic group; the check at a second representative, which this makes redundant, is off by default in the pipeline. The first factor is Shintani's formula for the Weil representation at a general element [9, Prop. 1.6], in the form given by Scheithauer [10, Thm. 4.7] and Strömberg [11]: for $\gamma$ with $c > 0$, $\rho_L(\gamma)e_\delta$ is a Gauss sum over $L/cL$ with the prefactor $\mathbf{e}\big((b^- - b^+)/8\big)\, c^{-n/2}\, |L^\vee/L|^{-1/2}$, $(b^+, b^-)$ the signature and $n = b^+ + b^-$, the same formula one gets by writing $\gamma\tau = a/c - 1/(c(c\tau + d))$ and applying Poisson summation to the inversion in the theta transformation formula [8, Thm. 4.1]. Two conventions enter. The input forms transform under the dual of Borcherds' $\rho_L$, its complex conjugate: the coset sum $\sum_w (f|w)\rho(w^{-1})e_0$ is well defined only if $\rho(Z^{-1})e_0 = \mathbf{e}(\frac{1}{4})e_0$ for $Z = S^2$, and at signature $(1, 2)$ that is the dual's sign ($\rho_L(Z)e_\gamma = \mathbf{e}(\frac{1}{4})e_{-\gamma}$ for Borcherds' $\rho_L$). And what is needed is $\rho(\gamma^{-1})$, whose lower-left entry is $-c < 0$; writing $\gamma^{-1} = Z^{-1} \cdot (-\gamma^{-1})$ and applying the formula to $-\gamma^{-1}$ gives the dual's prefactor $\mathbf{e}(-\frac{1}{8})$ times $\mathbf{e}(\frac{1}{4})$ from $Z^{-1}$. So, for the dual representation $\rho$ of a lattice of signature $(1, 2)$,

$$\rho(\gamma^{-1})e_0[\eta] = \mathbf{e}(\tfrac{1}{8})\, c^{-3/2}\, |L^\vee/L|^{-1/2}\, \mathbf{e}\Big(\frac{d\, Q(\eta)}{c}\Big) \sum_{\nu \in L/cL} \mathbf{e}\Big(\frac{a\, Q(\nu) + (\eta, \nu)}{c}\Big),$$

the Gauss sum factoring over the prime powers of $c$ (the quadratic term takes the cofactor, the linear term does not); the entry $d$ in the prefactor and $a$ in the sum are not interchangeable. The second factor is the Dedekind-eta multiplier system: for a monomial $\prod_d \eta(d\tau)^{r_d}$ of weight $\frac{1}{2}$, write $\left(\begin{smallmatrix} d & 0 \\ 0 & 1 \end{smallmatrix}\right)\gamma = g_d \left(\begin{smallmatrix} a_d & b_d \\ 0 & e_d \end{smallmatrix}\right)$ with $g_d \in SL_2(\mathbf{Z})$ of positive lower-left entry; then Apostol's transformation law $\eta(g\tau) = \varepsilon(g)\, (-i(c\tau + d))^{1/2}\eta(\tau)$, with $\varepsilon(g) = \exp\big(\pi i\big(\frac{a+d}{12c} + s(-d, c)\big)\big)$ a 24th root of unity given by a Dedekind sum, gives

$$\Big(\prod_d \eta(d\tau)^{r_d}\Big)\Big|\gamma = \mathbf{e}(-\tfrac{1}{8}) \prod_d \big[\varepsilon(g_d)\, \mathbf{e}\big(\tfrac{b_d}{24e_d}\big)\, e_d^{-1/2}\big]^{r_d}\, q^{\sum_d r_d a_d/(24e_d)} \prod_d \prod_{n \geq 1} \big(1 - \mathbf{e}(nb_d/e_d)\, q^{na_d/e_d}\big)^{r_d},$$

whose constant term is read off: the Borcherds inputs are holomorphic at every cusp other than $\infty$ and 0 and so are their monomials (lattice points of the polytope), so at the classes $1 < g < M$ the constant term is the product of the roots of unity when the order $\sum_d r_d a_d/e_d$ vanishes and 0 otherwise, and at the cusp 0 the representative $S$ makes the $q$-series rational. Every root of unity lies in $\mu_{24M}$ and the square roots are those of integers dividing $2M$, so the whole computation takes place in $\mathbf{Q}(\zeta_{24M})$ — in practice in the group ring $\mathbf{Q}[x]/(x^{24M} - 1)$ with the square roots as Gauss sums, reduced modulo the cyclotomic polynomial at the end, where the result must be a rational number. This is what the routine `MOMultipliersAlgebraic` does, and what the pipeline now uses. It reproduces the measured ground truth on $X_0^{15}(2)$ and the numerical coset evaluation on $X_0^{21}(2)$, $X_0^{10}(3)$, $X_0^{22}(3)$, $X_0^{34}(11)$, on the four probe eta quotients of Remark 12 (all three classes), and on the Borcherds inputs of the three composite-level bases $X_0^6(35)$, $X_0^{10}(21)$, $X_0^{14}(15)$ that have since been built, in seconds to minutes where the numerical route took an hour or more.

**Remark 14** (the conductor-prime terms at a squarefree level)**.** Proposition 19 is stated for prime $N$, and the code applies it prime by prime at a squarefree $N$. The statement extends with the same proof: the sum over $\nu \in L_-^\vee/L_-$ is regrouped by support, a coset supported at a single prime $p$ (a prime of the conductor, or a level prime not dividing $d_0$) has $E_0(\tau, s; \varphi_\nu) = \Phi(s)W_{0,p}(s, \varphi_\nu)/W_{0,p}(s, \varphi_0)$ because at every other prime $\nu_q = 0$ and the local factor is the zero coset's, and a coset supported at two or more primes has a product of vanishing ratios, hence a double zero and $\kappa_\nu^-(0) = 0$, exactly as in Proposition 10. So the restored terms are the sum over the primes of $N$ and of $f$ of the single-prime terms: Proposition 6's at a level prime not dividing $f$ (Remark 8), Proposition 19's fibre sum at a prime of $f$, with the local factors of Lemmas 20–23 at that prime alone. The code's fibre sum at $p$ enumerates the $p$-primary cosets, which is this regrouping.

**Non-fundamental discriminants.** Everything above assumes $d$ fundamental. Write $d = f^2 d_0$ with $d_0$ fundamental and $f > 1$ the conductor, $R_f = \mathbb{Z} + f\mathcal{O}_{d_0}$, and let $\lambda = \varphi(\sqrt{d})$ for an optimal embedding $\varphi$ of $R_f$, so that $\langle \lambda, \lambda \rangle = 2|d|$ as before. At a level prime $N$ dividing $f$ the lattice is different, and so is the bookkeeping; what is argued below — like everything in this note, a proposed argument until all of us have read it — is the general form of the restored terms (Proposition 19) and the local factors that enter it at every prime of the conductor (Lemmas 17, 18, 20, 21, 22, 23).

**Lemma 15** (the plane's isotropy is a property of the field)**.** *For every prime $p$, the quadratic space $L_- \otimes \mathbb{Q}_p = \lambda^\perp \subset B_0 \otimes \mathbb{Q}_p$ is isotropic if and only if $p$ splits in $k = \mathbb{Q}(\sqrt{d_0})$. In particular this does not depend on the conductor.*

*Proof.* Write $B = k \oplus kj$ with $j$ anticommuting with $k$ and $j^2 = \beta \in \mathbb{Q}^\times$, and $B_0$ for the trace-zero part. Then $\lambda \in k_0 = \mathbb{Q}\sqrt{d_0}$ and $\lambda^\perp = kj$: for $x \in k$, $\operatorname{tr}(\lambda \overline{xj}) = \operatorname{tr}(\lambda\, \bar{j}\, \bar{x}) = -\operatorname{tr}(\lambda \bar{x} j) = 0$ since $\lambda \bar{x} \in k$ and $\operatorname{tr}(yj) = 0$ for $y \in k$. The quadratic form is $Q(xj) = \operatorname{nrd}(xj) = -\beta\, N_{k/\mathbb{Q}}(x)$. So $\lambda^\perp \otimes \mathbb{Q}_p \cong (k_p, -\beta N)$, which represents 0 nontrivially exactly when $N_{k_p/\mathbb{Q}_p}$ does, i.e. when $k_p$ is not a field. □

**Lemma 16** (pole or no pole at the zero coset)**.** *In the counting normalisation of* Lemma lem:m0loc of the preprint*, $W_{0,p}(s, \varphi_0) = (1 - X)\sum_{k \geq 0} \alpha_k X^k$ with $X = p^{-s}$, $\alpha_0 = 1$ and $\alpha_k = p^{-k}\#\{x \in L_{-,p}/p^k L_{-,p} : Q(x) \equiv 0 \bmod p^k\}$. If $L_{-,p} \otimes \mathbb{Q}_p$ is anisotropic, the $\alpha_k$ are bounded and $W_{0,p}(s, \varphi_0)$ is holomorphic at $s = 0$. If it is isotropic, $\alpha_k \geq ck$ for some $c > 0$ and $W_{0,p}(s, \varphi_0)$ has a pole at $s = 0$.*

*Proof.* Anisotropic: $Q$ is a nonzero anisotropic form on $\mathbb{Z}_p^2$, so there is $c_0$ with $\operatorname{ord}_p Q(x) \leq c_0 + 2\min_i \operatorname{ord}_p(x_i)$ for all $x \neq 0$ (on the compact set of primitive $x$, $\operatorname{ord}_p Q(x)$ is bounded, and $Q(p^a x) = p^{2a} Q(x)$). Hence $Q(x) \equiv 0 \bmod p^k$ forces every coordinate of $x$ to vanish modulo $p^{\lceil (k - c_0)/2 \rceil}$, which leaves at most $p^{k + c_0}$ classes modulo $p^k$, and $\alpha_k \leq p^{c_0}$. Then $|(1 - X)\sum \alpha_k X^k| \leq p^{c_0}$ for $0 < X < 1$, so there is no pole. Isotropic: choose a primitive isotropic $e \in L_{-,p}$ and complete it to a basis $e, f'$, so $Q(xe + yf') = uxy + wy^2$ with $u = \langle e, f' \rangle \neq 0$. For each $0 \leq j \leq k - \operatorname{ord}_p u - 1$, the classes $y$ with $\operatorname{ord}_p y = j$ number $(p - 1)p^{k-j-1}$, and for each of them $Q \equiv 0 \bmod p^k$ is a congruence on $x$ modulo $p^{k-j-\operatorname{ord}_p u}$ with $p^{j + \operatorname{ord}_p u}$ solutions modulo $p^k$; summing, $\alpha_k \geq (k - \operatorname{ord}_p u)(p - 1)/p$, which grows linearly, and the pole follows since $(1 - X)\sum \alpha_k X^k \geq (1 - X)\sum_{k \leq K} ckX^k$ for every $K$. □

**Lemma 17** (the lattice at a conductor prime)**.** *Let $p$ be an odd level prime, $p \,\|\, N$, $p \nmid d_0$, $p^k \,\|\, f$ with $k \geq 1$, and $\lambda$ the image of $\sqrt{d} = f\sqrt{d_0}$ under an optimal embedding of $R_f$. In the model $\mathcal{O} \otimes \mathbb{Z}_p = \begin{pmatrix} \mathbb{Z}_p & \mathbb{Z}_p \\ p\mathbb{Z}_p & \mathbb{Z}_p \end{pmatrix}$, $L_p = \{\begin{pmatrix} a & b \\ pc & -a \end{pmatrix}\}$ with $Q = \operatorname{nrd}$, so $Q(a, b, c) = -a^2 - pbc$:*

(1) $\lambda = \begin{pmatrix} pa' & b \\ p^2c' & -pa' \end{pmatrix}$ *with* $b$ *a unit and* $a'^2 + bc' = -|d|/p^2$;
(2) $L_{-,p}$ *has the basis* $e_1 = (1, 0, -2a'/b)$, $e_2 = (0, 1, -pc'/b)$ *and, after the integral change of basis* $e_1' = e_1$, $e_2' = e_2 - (pa'/b)e_1$, *the Gram form* $Q = -\alpha^2 - p^{2k}|d_0|v^2\beta^2$ *with* $v$ *a unit: a unit line orthogonal to a* $p^{2k}$*-scaled line;*
(3) $[L_p : \mathbb{Z}_p\lambda \oplus L_{-,p}] = p^{2k-1}$;
(4) $L_{-,p}^\vee/L_{-,p} \cong \mathbb{Z}/p^{2k}$, *generated by* $e_2'/p^{2k}$; *its cosets of integral norm are the* $p^k - 1$ *multiples of* $e_2'/p^k$, *and of these exactly the* $p - 1$ *multiples of* $e_2'/p$ *lie in* $L_p^\vee$. *In particular the direct-sum decomposition of Lemma 1 fails, and a coset of* $L_{-,p}^\vee/L_{-,p}$ *enters the decomposition of some* $\varphi_\eta$, $\eta \in L^\vee/L$, *only if it is one of those* $p - 1$; *the others can pair only with* $x \notin L^\vee$, $x + \nu \in L^\vee$ *(Proposition 19).*

*Proof.* Write $\lambda = (a, b, c)$. Then $Q(\lambda) = -a^2 - pbc = |d| \cdot (\text{unit})^2$ is divisible by $p^{2k}$ with $2k \geq 2$, so $p \mid a$. Optimality of the embedding of $R_f$ says $\lambda/p \notin \mathcal{O} \otimes \mathbb{Z}_p$, i.e. not all of $a, b, c$ are divisible by $p$, so $b$ or $c$ is a unit. Conjugation by the Atkin–Lehner element $w_p = \begin{pmatrix} 0 & 1 \\ p & 0 \end{pmatrix}$ normalises $\mathcal{O} \otimes \mathbb{Z}_p$, preserves the norm, and sends $(a, b, c)$ to $(-a, c, b)$; since everything below is an isometry invariant of the pair $(L_p, \lambda)$, we may take $b$ to be the unit. Then $p^2 \mid a^2$ forces $p \mid c$, which is (1). Conjugating further by $\begin{pmatrix} 1 & 0 \\ pt & 1 \end{pmatrix} \in (\mathcal{O} \otimes \mathbb{Z}_p)^\times$ replaces $a'$ by $a' - tb$, so we may and do take $a' = 0$ below; this is what makes the pairing $\langle e_2', x\rangle = -p\gamma + (p^2c'/b)\beta$ free of an $\alpha$-term in the proof of (4). For $x = (\alpha, \beta, \gamma) \in L_p$, $\langle x, \lambda\rangle = -\operatorname{tr}(x\lambda) = -(2\alpha pa' + \beta p^2c' + \gamma pb)$, so $x \perp \lambda$ iff $\gamma = -(2a'\alpha + pc'\beta)/b$, which gives the basis in (2); substituting, $Q|_{L_{-,p}} = -\alpha^2 + (2pa'/b)\alpha\beta + (p^2c'/b)\beta^2$, and completing the square gives $-\alpha'^2 + (p^2/b^2)(a'^2 + bc')\beta^2 = -\alpha'^2 - p^{2k}|d_0|v^2\beta^2$ by (1). For (3) the determinant of the matrix with rows $\lambda, e_1, e_2$ is $(2p/b)(a'^2 + bc')$, of valuation $2k - 1$. For (4), $\langle e_2', e_2'\rangle = -2p^{2k}|d_0|v^2$ and $\langle e_1', e_1'\rangle = -2$ are the only nonzero pairings, so the dual is spanned by $e_1'/2$ and $e_2'/(2p^{2k}|d_0|v^2)$, i.e. by $e_2'/p^{2k}$ up to units; $Q(je_2'/p^{2k}) = -|d_0|v^2j^2/p^{2k}$ is integral iff $p^k \mid j$. Finally $\langle e_2'/p^m, x\rangle = -(1/p^{m-1})(\gamma + (pc'/b)\beta) + \dots$ is integral on $L_p$ iff $m \leq 1$, which leaves the multiples of $e_2'/p$. □

**Lemma 18** (the lattice at the conductor prime 2)**.** *Let* $2 \parallel N$, $2 \nmid d_0$, $2^k \parallel f$ *with* $k \geq 1$, *and* $\lambda$ *the image of* $\sqrt{d}$ *under an optimal embedding of* $R_f$, *in the model* $\mathcal{O} \otimes \mathbb{Z}_2 = \begin{pmatrix} \mathbb{Z}_2 & \mathbb{Z}_2 \\ 2\mathbb{Z}_2 & \mathbb{Z}_2 \end{pmatrix}$ *with* $Q = \text{nrd}$, $Q(a, b, c) = -a^2 - 2bc$ *for* $\begin{pmatrix} a & b \\ 2c & -a \end{pmatrix}$.

(1) *After conjugating by* $w_2$ *if necessary,* $\lambda = (2a', 2b', 2c')$ *with* $b'$ *a unit,* $c'$ *even and* $a'^2 + 2b'c' = -4^{k-1}|d_0|$ *(*$a'$ *odd exactly when* $k = 1$*);* $\lambda_0 = \lambda/2$ *is primitive in* $L_2$ *and* $L_{+,2} = \mathbb{Z}_2\lambda_0$.
(2) $L_{-,2}$ *has the basis* $e_1 = (1, 0, -a'/b')$, $e_2' = (0, 1, -c'/b') - (a'/b')e_1$ *with* $Q = -\alpha^2 - 4^{k-1}c\,\beta^2$, $c = |d_0|/b'^2 \equiv 3 \bmod 4$, *and* $-c \equiv 1 \bmod 8$ *exactly when* 2 *splits in* $k$.
(3) $[L_2 : L_{+,2} \oplus L_{-,2}] = 2^{2k-2}$; *in particular the sum is direct at conductor* 2.
(4) $L_{-,2}^\vee/L_{-,2} \cong \mathbb{Z}/2 \times \mathbb{Z}/2^{2k-1}$, *generated by* $e_1/2$ *and* $e_2'/2^{2k-1}$. *Its* $2^k$ *cosets of integral norm are* $(0, t/2^{k-1})$, $t \bmod 2^{k-1}$, *and* $(1/2, t/2^k)$, $t$ *odd mod* $2^k$; *writing* $2^{k-\rho}$ *for the order of a nonzero one,* $\rho = 1 + \operatorname{ord}_2 t \in [1, k-1]$ *for the first kind and* $\rho = 0$ *for the second, so there are* $2^{k-\rho-1}$ *cosets of each* $\rho \in [0, k-1]$. *A nonzero coset lies in* $L_2^\vee$ *iff it has order* 2 *(*$\rho = k - 1$*); there is exactly one.*

*Proof.* Over $\mathbb{Z}_2$, $R_f \otimes \mathbb{Z}_2 = \mathbb{Z}_2 + 2^k\mathcal{O}_k$ is generated by $2^{k-1}(1 + \sqrt{d_0})$, whose image is $2^{k-1} + \lambda/(2f')$ with $f = 2^kf'$, $f'$ a unit; it lies in $\mathcal{O} \otimes \mathbb{Z}_2$ iff $\lambda/2$ does, i.e. iff $a, b, c$ are all even.

Optimality says the order of conductor $2^{k-1}$ does not embed: for $k \geq 2$ its generator $2^{k-2}(1+\sqrt{d_0})$ maps to $2^{k-2} + \lambda/(4f')$, so $\lambda/4 \notin \mathcal{O} \otimes \mathbb{Z}_2$, i.e. not all of $a', b', c'$ are even; for $k = 1$ the generator $(1+\sqrt{d_0})/2$ maps to $(1+\lambda/(2f'))/2$, which lies in the order iff $a'$ is odd and $b', c'$ are even. From $Q(\lambda) = 4^k|d_0|$ we get $a'^2 + 2b'c' = -4^{k-1}|d_0|$. If $k \geq 2$ the right side is $\equiv 0 \bmod 4$, which forces $a'$ even and then $b'c'$ even, so by optimality exactly one of $b', c'$ is odd; if $k = 1$ the right side is odd, so $a'$ is odd, $2b'c' = -|d_0| - a'^2 \equiv 0 \bmod 4$ makes $b'c'$ even, and optimality again says one of $b', c'$ is odd. Conjugation by $w_2$ sends $(a', b', c')$ to $(-a', c', b')$, so we may take $b'$ odd and $c'$ even. Then $\lambda_0 = (a', b', 2c')$ has an odd entry, hence is primitive. This is (1).

For (2): $\langle x, \lambda\rangle = -2(a\alpha + c\beta + b\gamma) = -4(a'\alpha + c'\beta + b'\gamma)$ for $x = (\alpha, \beta, \gamma)$, so $x \perp \lambda$ iff $\gamma = -(a'\alpha + c'\beta)/b'$, which gives the basis $e_1, e_2 = (0, 1, -c'/b')$ with $Q(\alpha e_1 + \beta e_2) = -\alpha^2 + (2a'/b')\alpha\beta + (2c'/b')\beta^2$; completing the square (an integral change of basis since $b'$ is a unit) gives $-\alpha'^2 + ((a'^2 + 2b'c')/b'^2)\beta^2 = -\alpha'^2 - 4^{k-1}c\,\beta^2$. As $b'$ is odd, $c \equiv |d_0| \bmod 8$, and $d_0 \equiv 1 \bmod 4$ gives $|d_0| \equiv 3 \bmod 4$, with $|d_0| \equiv 7 \bmod 8$ iff $d_0 \equiv 1 \bmod 8$ iff 2 splits.

For (3): the Gram matrix of $\langle\cdot,\cdot\rangle = 2Q$ on $L_2$ in the coordinates $(\alpha, \beta, \gamma)$ is $(-2) \oplus \begin{pmatrix} 0 & -2 \\ -2 & 0 \end{pmatrix}$, of determinant 8; on $L_{+,2} \oplus L_{-,2}$ it is $\mathrm{diag}(2 \cdot 4^{k-1}|d_0|, -2, -2 \cdot 4^{k-1}c)$, of determinant $8 \cdot 4^{2k-2}\cdot$unit. The ratio of determinants is the square of the index.

For (4): the dual of $\mathrm{diag}(-2, -2^{2k-1}c)$ is spanned by $e_1/2$ and $e_2'/(2^{2k-1}c)$, and $Q(se_1/2 + te_2'/2^{2k-1}) = -s^2/4 - ct^2/2^{2k}$. For $s = 0$ this is integral iff $2^k \mid t$; for $s = 1$ iff $ct^2 \equiv -2^{2k-2} \bmod 2^{2k}$, i.e. $t = 2^{k-1}t_1$ with $t_1$ odd and $ct_1^2 \equiv -1 \bmod 4$, which holds since $c \equiv 3 \bmod 4$. The orders are read off the denominators. Finally $\langle \nu, x\rangle = -2(\nu_\alpha\alpha + \nu_\beta\gamma + \nu_\gamma\beta)$, so $\nu \in L_2^\vee$ iff $2\nu_\alpha, 2\nu_\beta, 2\nu_\gamma \in \mathbb{Z}_2$. For $\nu = ue_1 + we_2'$ one has $\nu_\beta = w$, so $2w \in \mathbb{Z}_2$ is necessary: for the first kind $w = t/2^{k-1}$ this forces $2^{k-2} \mid t$, i.e. $\rho = k - 1$, and for the second kind $w = t/2^k$ with $t$ odd it forces $k = 1$, again $\rho = k - 1$; in both cases $2\nu_\alpha = 2u - 2(a'/b')w$ and $2\nu_\gamma = -2(a'/b')u + 2w(a'^2/b'^2 - c'/b')$ are then integral. □

So at a level prime dividing the conductor the picture splits by the field: if $p$ splits, the plane is isotropic and the zero coset has its pole; if $p$ is inert, the plane is anisotropic and there is no pole — a case only non-fundamental $d$ can produce, since for fundamental $d$ an inert $p \,\|\, N$ admits no optimal embedding (Eichler's condition $\nu_p = 1 + (\frac{d}{p}) = 0$), whereas for $p \mid f$ the local embedding number is 2 regardless. In neither case does the argument of Proposition 6 apply as it stands, because it identifies a coset of $L_-^\vee/L_-$ with a coset of $L^\vee/L$ through $L_N = L_{+,N} \oplus L_{-,N}$, and Lemma 17(3) says the sum is not direct. What survives is the bookkeeping before that identification.

**Proposition 19** (the restored terms without the direct-sum hypothesis)**.** *Let $N$ be prime and $d < 0$ any discriminant with $\mathrm{CM}(d) \neq \varnothing$. The terms of* [5, Thm. B] *in which $\kappa^-$ is evaluated at 0 at a nonzero coset total, per CM point,*

$$-\frac{1}{4} \sum_{\substack{\nu \in L_-^\vee/L_- \\ \nu \neq 0}} \kappa_\nu^-(0) \sum_{\substack{x \in L_+^\vee \\ x+\nu \in L^\vee}} c_{[x+\nu]}\big(-Q(x)\big),$$

*where $\kappa_\nu^-(0) = 0$ unless $Q(\nu) \in \mathbb{Z}$. If moreover $\nu$ is supported at the single prime $p$ (i.e. $\nu_q = 0$ for $q \neq p$), then with $X = p^{-s}$ and $W_{0,p}(s, \varphi_\nu)$ the local factor at $p$ in any fixed normalisation,*

$$\kappa_\nu^-(0) = \log p \cdot \frac{d}{dX}\Big[\frac{W_{0,p}(s, \varphi_\nu)}{W_{0,p}(s, \varphi_0)}\Big]_{X=1},$$

*the ratio being a rational function of $X$ that vanishes at $X = 1$.*

*Proof.* The display is the first display in the proof of Proposition 6, which is obtained from Theorem B by regrouping alone, before $L_N = L_{+,N} \oplus L_{-,N}$ is used; the prefactor is Theorem B's $-\frac{1}{4}$ per point. For $Q(\nu) \notin \mathbb{Z}$ some $p$ has $Q(\nu_p) \notin \mathbb{Z}_p$, and then $W_{0,p}(s,\varphi_\nu) \equiv 0$ ([3, Thm. 4.3(i), 4.4(i)]: every $x \in \nu_p + L_{-,p}$ has $Q(x) \equiv Q(\nu_p) \notin \mathbb{Z}_p$), so $E_0(\tau,s;\varphi_\nu) \equiv 0$. For $\nu$ supported at $p$, exactly as in the proof of Proposition 4, $E_0(\tau,s;\varphi_\nu) = \Phi(s)\, W_{0,p}(s,\varphi_\nu)/W_{0,p}(s,\varphi_0)$ with $\Phi(0) = -1$; the series $E(\tau,s;\varphi_\nu)$ is incoherent, hence vanishes at $s=0$, so the ratio vanishes at $X=1$, and $dX/ds = -X\log p$ gives the formula. The two facts used here for the plane of a non-maximal order — holomorphy of $E(\tau,s;\varphi)$ at $s=0$ [3, Cor. 2.5] and its vanishing there — are properties of the quadratic space $L_- \otimes \mathbb{Q} \cong (k, -cN_{k/\mathbb{Q}})$ and of $\varphi \in S(V(\mathbb{A}_f))$, not of the lattice: [3, Cor. 2.5] is stated for an arbitrary Schwartz function, and the series is incoherent because its finite components come from $(V,Q)$ and the archimedean one from $(V,-Q)$ [6, after (2.18)], whatever lattice $\varphi$ is the characteristic function of. Bruinier–Ehlen–Yang run this argument for binary lattices of arbitrary discriminant, the orders $\mathcal{O}_D$ included [7, §3]. (For $\nu$ supported at two or more primes the product of the vanishing ratios has a double zero and $\kappa_\nu^-(0) = 0$, as in Proposition 10.) □

At a conductor prime the inner sum is genuinely a sum over a fibre. Two kinds of pairs $(x,\nu)$ appear that do not exist for fundamental $d$: a coset $\nu \in L^\vee$ pairs with $x \notin L$ such that $x+\nu \in L$ (so $[x+\nu]=0$ and the coefficient is $c_0(-Q(x))$, which includes the pole of $f$ at $\infty$), and a coset $\nu \notin L^\vee$ pairs with $x \notin L^\vee$ such that $x+\nu \in L^\vee$. The vectors $x$ that occur are the vectors $\lambda_0/p^j$ of the CM line, of norm $Q(\lambda_0)/p^{2j}$: the primitive CM vectors of the discriminants $d/p^{2j}$. So the correction at a conductor prime sees the poles of $f$ at the CM points of the *smaller* discriminants $d/p^2, d/p^4, \dots$, which no $d$-independent multiplier can.

Computation on $X_0^{15}(2)$. Everything in Proposition 19 is computable from the lattice: the integral cosets of $L_-^\vee/L_-$ supported at 2, the fibre over each, the coefficients $c_{[x+\nu]}(-Q(x))$ (the cusp-0 principal part of $f$ at the nonzero cosets by Lemma 7, $c_\eta(0) = 2\,\mathrm{mult}(f)$ at a nonzero isotropic coset, and the cusp-$\infty$ principal part added at $\eta=0$), and the local factors $W_{0,2}(s,\varphi_\nu) = (1-X)\sum_k \alpha_k X^k$ by counting on $L_- = L \cap \lambda^\perp$ as in Lemma 16, reconstructed as rational functions from $\alpha_0,\dots,\alpha_{10}$ (`level-p2/fibresum.m` on the campaign branch; every reconstruction has at least two spare terms, and the results are the series of Lemma 20 below). With $\lambda_0 = \lambda/2$ ($Q(\lambda_0) = |d|/4$) and $\nu_0$ the integral coset lying in $L^\vee$:

| $d$ | $f$, type | $W_{0,2}(s,\varphi_0)$ | integral cosets $\nu$: $W_{0,2}(s,\varphi_\nu)$, $\kappa_\nu^-(0)/\log 2$ | fibre over $\nu$ ($Q(x$ |
|---|---|---|---|---|
| $-15,-7$ | 1, split | $1/(1-X)$ | two, in $L^\vee$: $1$, $-1$ | $x=0$ |
| $-60,-28$ | 2, split | $(1-X+X^2)/(1-X)$ | $\nu_0$: $1+X$, $-2$ | $x=0$ |
| $-240$ | 4, split | $(1-X+X^2-X^3+2X^4)/(1-X)$ | $\nu_0$: $1+X^2+2X^3$, $-2$ | $x=0$; $x=\pm\lambda_0/2$ |
| | | | $\pm\nu_1 \notin L^\vee$: $1+X$, $-1$ | none below the p |
| $-12$ | 2, inert | $(1+X+X^2)/(1+X)$ | $\nu_0$: $1-X$, $-\frac{2}{3}$ | $x=0$; $x=\pm\lambda_0/2$ |
| $-48$ | 4, inert | $(1+X+X^2+X^3+2X^4)/(1+X)$ | $\nu_0$: $(1-X)(1+X+2X^2)$, $-\frac{4}{3}$ | $x=0$; $x=\pm\lambda_0/2$ |
| | | | $\pm\nu_1 \notin L^\vee$: $1-X$, $-\frac{1}{3}$ | $x=\pm\lambda_0/4$, $[x+$ |

(The fibres are listed up to the largest pole of the nine forms of the model set, $q^{-30}$ at $\infty$ and $q^{-3/4}$ at the cusp 0.) Writing $m = \frac{1}{2}c_\eta(0)$ for the fundamental multiplier, $a = c_\infty(-|d|/16)$ for the coefficient of $f$ at the pole exponent $Q(\lambda_0/2)$ and $b$ for the coefficient of $q^{-3/4}$ in $f|S$, the proposition gives per point

$$d=-240: \quad (m+a)\log 2, \qquad\qquad d=-48: \quad \left(\tfrac{2}{3}(m+a)+\tfrac{1}{3}b\right)\log 2,$$

while at $d=-60,-28$ it gives $m\log 2$ (one coset at twice the weight of the two fundamental ones) and at $d=-15,-7$ the $m\log 2$ of Proposition 6. These predictions were compared, for all nine forms, with the values forced by Table 45: each form is a function on the genus-0

curve $X_0^{15}(2)/W$ whose divisor is known (Lemma 25 of [5]), so its value at every CM point is $C\prod_i|s(d)-s(d_i)|^{m_i}$ with $s$ Guo–Yang's Hauptmodul; the constant $C$ was read off at $d=-7$ and the formula checked at six to eight further fundamental and conductor-2 points per form before being used at $-240$ and $-48$ (`level-p2/evidence.log`). The fibre sum reproduces all 18 values exactly. The rule "add $m\log p$ when $p$ splits, nothing when $p$ is inert", which the routine applied before the fibre sum, is right at $-240$ only for the five forms with $a=0$ (those whose divisor misses $\tau_{-60}$) and at $-48$ only for the three forms with $m+a=0$ and $b=0$ — among them the form on which the rule had been tested — and Yang's conductor term for the pairs with $Q(x)=m$, which the routine also carried, gives the irrational $2^{4/3}$ at $-48$ for the two forms with $b\neq 0$ (those pairs now enter only through the fibre sum). The vanishing at $-48$ for the tested form is thus the identity $c_\eta(0)+2c_\infty(-3)=0$ for that form, not the absence of a term.

The local factors in the table are instances of a closed form.

**Lemma 20** (the $m=0$ local factors at a conductor prime)**.** *Let $p\,\|\,N$, $p\nmid d_0$, $p^k\,\|\,f$ with $k\geq 1$, and $\epsilon=(\frac{d_0}{p})=\pm1$, with $L_{-,p}$ as in Lemma 17 ($p$ odd) or Lemma 18 ($p=2$). In the counting normalisation $W_{0,p}(s,\varphi_\nu)=(1-X)\sum_{j\geq0}\alpha_j(\nu)X^j$, $X=p^{-s}$, $\alpha_j(\nu)=p^{-j}\#\{x\in(\nu+L_{-,p})/p^jL_{-,p}:Q(x)\equiv0 \bmod p^j\}$:*

(1) *For $p$ odd the cosets of integral norm are $\nu_r=(0,r/p^k)$, $r \bmod p^k$; for $r\neq0$ let $\rho=\operatorname{ord}_p r<k$, so that $\nu_r$ has order $p^{k-\rho}$ and lies in $L_p^\vee$ iff $\rho=k-1$. For $p=2$ let $\nu_r$ run over the $2^k-1$ nonzero integral cosets of Lemma 18(4), with $\rho$ defined there by the order $2^{k-\rho}$. In both cases $\alpha_j(\nu_r)=p^{\lfloor j/2\rfloor}$ for $j\leq2\rho$ and $\alpha_j(\nu_r)=(1+\epsilon)p^\rho$ for $j>2\rho$, i.e.*

$$W_{0,p}(s,\varphi_{\nu_r})=(1-X)\sum_{j=0}^{2\rho}p^{\lfloor j/2\rfloor}X^j+(1+\epsilon)\,p^\rho X^{2\rho+1}.$$

(2) *For the zero coset, $\alpha_j(0)=p^{\lfloor j/2\rfloor}$ for $j\leq2k$ and $\alpha_j(0)=(1+\epsilon)(p-1)p^{k-1}\lceil(j-2k)/2\rceil+p^{k-(j \bmod 2)}$ for $j>2k$, i.e.*

$$W_{0,p}(s,\varphi_0)=(1-X)\Big[\sum_{j=0}^{2k}p^{\lfloor j/2\rfloor}X^j+\frac{X^{2k+1}(p^{k-1}+p^kX)}{1-X^2}\Big]+\frac{(1+\epsilon)(p-1)p^{k-1}X^{2k+1}}{1-X^2}.$$

(3) *Consequently $\kappa_{\nu_r}^-(0)=-\dfrac{2p^{\rho-k+1}}{p-1}\log p$ if $p$ splits, and*

$$\kappa_{\nu_r}^-(0)=-\frac{2}{p^{k-1}(p+1)}\Big(p^\rho+\frac{2(p^\rho-1)}{p-1}\Big)\log p$$

*if $p$ is inert; all other nonzero cosets have $\kappa_\nu^-(0)=0$.*

*Proof.* With $Q(u,t)=-(u^2+cp^{2k}t^2)$ the dual is generated by $e_2/p^{2k}$, and $Q(0,r/p^{2k})=-cr^2/p^{2k}$ is integral iff $p^k\mid r$, which is (1)'s list and Lemma 17(4). A vector of $\nu_r+L_{-,p}$ is $(u,w)$ with $w=r+p^kt\in r+p^k\mathbb{Z}_p$, and for $r\neq0$ every such $w$ has $\operatorname{ord}_p w=\rho$. The congruence $u^2+cw^2\equiv0 \bmod p^j$ reads $u^2\equiv0$ when $j\leq2\rho$, with $p^{\lfloor j/2\rfloor}$ solutions $u \bmod p^j$ for each of the $p^j$ values of $t$; when $j>2\rho$ it forces $u=p^\rho u'$ with $u'^2\equiv-c(w/p^\rho)^2 \bmod p^{j-2\rho}$, which has 2 solutions $u' \bmod p^{j-2\rho}$ if $-c$ is a square mod $p$ — i.e. $\epsilon=1$, since $-c=d_0v^2$ — and none otherwise, giving $(1+\epsilon)p^\rho$ values of $u \bmod p^j$ for each $t$. This is (1). For the zero coset, $w=p^kt$; if $t\equiv0 \bmod p^j$ or $2k+2\operatorname{ord}_p t\geq j$ the congruence is $u^2\equiv0$ with $p^{\lfloor j/2\rfloor}$ solutions,

and the $t$ with $2k+2v<j$, $v=\operatorname{ord}_p t$, number $(p-1)p^{j-v-1}$ and each admits $(1+\epsilon)p^{k+v}$ values of $u$ by the same Hensel count. Summing, for $j>2k$,

$$p^j\alpha_j(0)=(1+\epsilon)(p-1)p^{j+k-1}\left\lceil\frac{j-2k}{2}\right\rceil+p^{\lfloor j/2\rfloor}\Big(1+\sum_{v\ge\lceil(j-2k)/2\rceil}^{j-1}(p-1)p^{j-1-v}\Big)=(1+\epsilon)(p-1)p^{j+k-1}\left\lceil\frac{j-2k}{2}\right\rceil+p^{\lfloor j/2}$$

and $\lfloor j/2\rfloor-\lceil(j-2k)/2\rceil=k-(j \bmod 2)$; for $j\le 2k$ only $u^2\equiv 0$ remains. The generating functions follow from $\sum_{n\ge1}\lceil n/2\rceil X^n=X/((1-X)(1-X^2))$. For (3) apply Proposition 19. If $\epsilon=-1$, $W_{0,p}(s,\varphi_0)$ is finite at $X=1$ with value $(p^{k-1}+p^k)/2$ (only the middle term survives the factor $1-X$) and $W_{0,p}(s,\varphi_{\nu_r})=(1-X)\sum_{j\le2\rho}p^{\lfloor j/2\rfloor}X^j$ has derivative $-\sum_{j\le2\rho}p^{\lfloor j/2\rfloor}=-(p^\rho+2(p^\rho-1)/(p-1))$ there. If $\epsilon=1$, $(1-X)W_{0,p}(s,\varphi_0)\to(p-1)p^{k-1}$ at $X=1$ while $W_{0,p}(s,\varphi_{\nu_r})\to 2p^\rho$, so the ratio has derivative $-2p^\rho/((p-1)p^{k-1})$.

For $p=2$ the form is $u^2+4^{k-1}cw^2$ with $c\equiv3 \bmod 4$ and $-c\equiv1 \bmod 8$ iff $\epsilon=1$ (Lemma 18), and the count of $u'$ modulo $2^n$ with $u'^2\equiv -c\,w'^2$, $w'$ odd, is 1, 2, $2(1+\epsilon)$ for $n=1$, $n=2$, $n\ge3$. For the zero coset and for the cosets $(0,t/2^{k-1})$ write the condition as $u^2+cW^2\equiv0 \bmod 2^j$ with $W=2^{k-1}w$, so that $\operatorname{ord}_2 W=k-1+\operatorname{ord}_2 w$ for the zero coset and $\operatorname{ord}_2 W=\rho-1$ for the coset $(0,t/2^{k-1})$ of depth $\rho$. With $v=\operatorname{ord}_2 W$ and $n=j-2v$: for $n\le0$ the condition is $u^2\equiv0$, with $2^{\lfloor j/2\rfloor}$ solutions; for $n=1,2$ it is $u=2^vu'$ with 1, resp. 2 values of $u'$ modulo $2^n$, i.e. $2^v$, resp. $2^{v+1}$ values of $u$ modulo $2^j$, which again equals $2^{\lfloor j/2\rfloor}$ since $j=2v+1,2v+2$; for $n\ge3$ it is $(1+\epsilon)2^{v+1}$ values. These are exactly the per-$w$ counts of the odd case with $p^{k+v}$ replaced by $2^{(k-1+\operatorname{ord}_2 w)+1}$, so the sums are the same and (1), (2) hold verbatim: for the coset of depth $\rho\ge1$ the count is $2^{\lfloor j/2\rfloor}$ for $j\le2\rho$ and $(1+\epsilon)2^\rho$ beyond. For a coset $(1/2,t/2^k)$, $t$ odd, put $U=2u$ and $W=2^kw$, both odd; the condition becomes $U^2+cW^2\equiv0 \bmod 2^{j+2}$, which holds for every $U$ when $j=0$ (as $1+c\equiv0 \bmod 4$) and for $j\ge1$ has $2(1+\epsilon)$ solutions $U$ modulo $2^{j+2}$, i.e. $(1+\epsilon)$ solutions $u$ modulo $2^j$ for each of the $2^j$ values of $w$: $\alpha_j=1+\epsilon$ for $j\ge1$, the case $\rho=0$. Part (3) follows as before. □

The formulas were also checked by brute-force counting on the plane $L\cap\lambda^\perp$ of $X_0^{10}(3)$ at $p=3$ — $d=-72,-648$ (split, $k=1,2$) and $d=-387,-603$ (inert, $k=1$), every 3-primary integral coset, six terms each (`level-p2/wcond_check.m`). At $p=2$ the six rows of the table are the lemma's series for $k=1,2$ and both splitting types, and the factors $-2$, $-1$, $-\frac43$, $-\frac13$, $-\frac23\log2$ are the values of (3) at $(k,\rho)=(1,0),(2,1),(2,0)$ in the split case and $(2,1),(2,0),(1,0)$ in the inert case. With Lemma 20, Proposition 19 is explicit at a conductor prime: the restored terms there are

$$-\frac14\sum_{r\ne0}\kappa^-_{\nu_r}(0)\sum_{\substack{x\in L^\vee_+\\ x+\nu_r\in L^\vee}}c_{[x+\nu_r]}\big(-Q(x)\big),$$

with the fibre over each $\nu_r$ read off the lattice as in the computation above. The routine evaluates the sum as written at a level prime dividing the conductor (`MOFibreCorrection`), with the closed-form $\kappa^-_\nu(0)$ and the fibre read off the lattice, and reproduces the 18 values of the table together with the 28 at conductors 1 and 2 (`tests/MOPoleSum.m`).

**A prime of the conductor outside the level.** A prime $q$ dividing the conductor need not divide the level ($q\mid D$ is impossible: at a prime of $D$ the only order of $k_q$ embedding optimally into the maximal order of the division algebra is the maximal one). There the same thing happens with one difference, and it puts terms in $\log q$ into the formula at a prime where the lattice $L$ itself is unimodular.

**Lemma 21** (the lattice at an odd conductor prime outside the level)**.** *Let $q$ be odd, $q \nmid DN$, $q^k \parallel f$ with $k \geq 1$, and $\lambda$ the image of $\sqrt{d}$ under an optimal embedding of $R_f$. In $\mathcal{O} \otimes \mathbb{Z}_q = M_2(\mathbb{Z}_q)$, after conjugating by $GL_2(\mathbb{Z}_q)$, $\lambda = \begin{pmatrix} 0 & d \\ 1 & 0 \end{pmatrix}$, so $L_{+,q} = \mathbb{Z}_q\lambda$ and*

(1) *$L_{-,q} = \langle -1 \rangle \perp \langle -|d| \rangle$, with basis $e_1 = \mathrm{diag}(1,-1)$, $e_2 = \begin{pmatrix} 0 & -d \\ 1 & 0 \end{pmatrix}$; $[L_q : L_{+,q} \oplus L_{-,q}] = q^{2k}$;*

(2) *$L_{-,q}^\vee / L_{-,q}$ has $q$-part $\mathbb{Z}/q^{2k}$ (if $q \nmid d_0$) or $\mathbb{Z}/q^{2k+1}$ (if $q \mid d_0$), generated by $e_2/|d|$, and in both cases its cosets of integral norm are the $q^k$ cosets $\nu_r = (0, r/q^k)$, $r \bmod q^k$, of order $q^{k-\rho}$ with $\rho = \mathrm{ord}_q r$; none of the nonzero ones lies in $L_q^\vee = L_q$;*

(3) *the fibre over $\nu_r$ is $\{t\lambda : \ t \in -r/q^k + \mathbb{Z}_q\} \cap L_+^\vee$: the vectors $x = t\lambda$ with $t \equiv -r/q^k \bmod \mathbb{Z}_q$, whose norms $t^2|d|$ are those of the primitive CM vectors of the discriminants $d/q^{2(k-\rho)}$ and their multiples prime to $q$; the class $[x + \nu_r] \in L^\vee/L$ is trivial at $q$ and equals that of $x$ elsewhere, so for $x \in L[1/q]$ it is $0$ and the coefficient is $c_0(-Q(x))$;*

(4) *if $q \nmid d_0$ the plane is $\langle -1 \rangle \perp \langle -q^{2k} c \rangle$ with $c$ a unit and Lemma 20 applies verbatim; if $q \mid d_0$ it is $\langle -1 \rangle \perp \langle -q^{2k+1} c' \rangle$, anisotropic, with $\alpha_j(\nu_r) = q^{\lfloor j/2 \rfloor}$ for $j \leq 2\rho + 1$ and $0$ beyond, $\alpha_j(0) = q^{\lfloor j/2 \rfloor}$ for $j \leq 2k+1$ and $q^k$ beyond, hence*

$$\kappa_{\nu_r}^-(0) = -\frac{2(q^{\rho+1} - 1)}{(q-1)q^k} \log q.$$

*Consequently the restored terms at $q$ are, per CM point,*

$$-\frac{1}{4} \sum_{\rho=0}^{k-1} \kappa_\rho^-(0) \sum_{x \in q^{\rho-k}\Lambda \setminus q^{\rho-k+1}\Lambda} c_{[x]'}\big(-Q(x)\big), \qquad \Lambda = L^\vee \cap \mathbb{Q}\lambda,$$

*with $[x]'$ the class of $x$ away from $q$; when every such $x$ lies in $L[1/q]$ this is $-\frac{1}{2}\sum_{j=1}^{k} \kappa_{k-j}^-(0) \sum_{m \geq 1,\ q \nmid m} c_0\big(-m^2 Q$ a pole sum over the CM points of the lower discriminants $d/q^2, \ldots, d/q^{2k}$.*

*Proof.* Since $q^{2k} \mid Q(\lambda) = |d|$ and $\lambda$ is primitive (optimality: $\lambda/q \notin \mathcal{O}_q$, as $R_f \otimes \mathbb{Z}_q = \mathbb{Z}_q + q^k \mathcal{O}_{k,q}$ is generated by $q^k\sqrt{d_0} = \lambda/f'$ with $f'$ a unit), $\lambda$ reduces mod $q$ to a nonzero nilpotent matrix, of rank 1; a vector $v$ with $\lambda v \not\equiv 0$ then gives the basis $v, \lambda v$ of $\mathbb{Z}_q^2$, in which $\lambda = \begin{pmatrix} 0 & d \\ 1 & 0 \end{pmatrix}$. With $Q(a,b,c) = -a^2 - bc$ on $\begin{pmatrix} a & b \\ c & -a \end{pmatrix}$, $\langle x, \lambda \rangle = -(\beta + d\gamma)$, which gives the basis of $L_{-,q}$ and its Gram matrix; the determinants 2 of $L_q$ and $-8|d|^2$ of $L_{+,q} \oplus L_{-,q}$ give the index. The dual is generated by $e_2/|d|$, and with $|d| = q^{2k+\delta}c$ ($\delta \in \{0,1\}$, $c$ a unit) the $q$-part is generated by $e_2/q^{2k+\delta}$; $Q(te_2/q^{2k+\delta}) = -ct^2/q^{2k+\delta}$ is integral iff $q^{k+\delta} \mid t$, i.e. iff the coset is a multiple of $e_2/q^k$. No nonzero $\nu_r$ lies in $L_q$, which is $L_q^\vee$. For (3), $x = t\lambda = (0, td, t)$ and $\nu_r = (0, -dr/q^k, r/q^k)$, so $x + \nu_r \in L_q$ iff $t + r/q^k \in \mathbb{Z}_q$ (the other coordinate then differs from $d(t + r/q^k)$ by $2dr/q^k \in \mathbb{Z}_q$); at $p \neq q$ the condition is $x \in L_p^\vee$, and $\nu_r$ has no component there. For (4), the odd case is Lemma 20; in the ramified case write $W = q^k w$, so $Q = -(u^2 + qc'W^2)$ with $\mathrm{ord}_q(qc'W^2)$ odd, and $u^2 \equiv -qc'W^2 \bmod q^j$ has solutions only when both sides are $\equiv 0$, i.e. for $j \leq \mathrm{ord}_q(qc'W^2)$, with $q^{\lfloor j/2 \rfloor}$ of them; summing over $W$ as in the proof of Lemma 20 gives the series, which are bounded (no pole, as the plane is anisotropic), and Proposition 19 gives $\kappa^- = -\sum_{j \leq 2\rho+1} q^{\lfloor j/2 \rfloor}/q^k \cdot \log q$. The display is Proposition 19 with the fibres of (3), the cosets of depth $\rho$ covering the residues $t \equiv -r/q^k$ with $\mathrm{ord}_q r = \rho$ exactly once. □

Check. Table 45 does not reach such points, but the forms must still be functions of the one Hauptmodul $s$, so at $d = -588 = 14^2 \cdot (-3)$ on $X_0^{15}(2)$ (7 splits in $\mathbb{Q}(\sqrt{-3})$, three star points) the nine values (now sums over the three points) determine $N(s)$, $N(s-2)$ and $N((s+\frac{1}{12})(s-\frac{5}{4}))$, and a monic cubic $H \in \mathbb{Q}[X]$ with these absolute values at 0, 2 and (as a product) at $-\frac{1}{12}, \frac{5}{4}$ must exist — a rationality condition on the discriminant of a quadratic. With the term of Lemma 21 (per point $\frac{1}{6}c_\infty(-3)\log 7$, the pole of $f$ at $\tau_{-12}$) it exists,

$$H(X) = X^3 - \tfrac{191}{54}X^2 + \tfrac{343}{432}X - \tfrac{83^2}{2^8 \cdot 3^3 \cdot 7},$$

and the cubic field it cuts out has discriminant $-588$ and the same reduced polynomial $X^3 - X^2 + 5X + 1$ as the field of definition of the CM points computed by class field theory (`FieldsOfDefinitionOfCMPoint`); without the term no rational cubic exists. The 7-adic Newton polygon of $H$ puts every root at $\operatorname{ord}_7 = -\frac{1}{3}$: all three points reduce to the pole $\tau_{-12}$, which is the congruence $s(\tau_{dq^2}) \equiv s(\tau_d)$ at the primes above $q$ that the cyclic $q$-isogeny between the CM points predicts. The same happens at $d = -1960 = 7^2 \cdot (-40)$ (field discriminant $-1960$, roots divisible by 7 as $s(\tau_{-40}) = 0$), and at $d = -735 = 7^2 \cdot (-15)$, 7 inert, the term puts the factor 7 into $N(s - \frac{5}{4})$ as $s(\tau_{-15}) = \frac{5}{4}$ requires (`level-p2/qprime.m`, `classpoly.py`, `fod.m`). Three published values do reach such points, once they are forced into the table (the pipeline's candidate search offers conductors 1 and 2 only): on $X_0^{58}(1)$ and $X_0^{55}(1)$ — no level prime, so no other $m = 0$ term at all — [5] give $s(-27) = 3/19$ and 0, with $-27 = 3^2 \cdot (-3)$ and $3 \mid d_0$, the ramified case of (4); on $X_0^{21}(2)$ they give $s(-100) = 1/5$, with $-100 = 5^2 \cdot (-4)$ and 5 split in $\mathbb{Q}(i)$, the case of Lemma 20. The terms are $\log 3$, $-\log 3$ and $-4\log 5$ on the Hauptmodul form, and with them the pipeline reproduces all three published values (by cross-ratio against three other points of each table, as in `tests/_offline/GuoYangCheck.m`; `level-p2/gyforce_*.m`).

**Lemma 22** (the conductor prime 2 outside the level)**.** *Let $2 \nmid DN$, $2^k \parallel f$ with $k \geq 1$, and $\lambda$ the image of $\sqrt{d}$ under an optimal embedding of $R_f$. In $\mathcal{O} \otimes \mathbb{Z}_2 = M_2(\mathbb{Z}_2)$, after conjugating by $GL_2(\mathbb{Z}_2)$, $\lambda_0 = \lambda/2 = \begin{pmatrix} 0 & d/4 \\ 1 & 0 \end{pmatrix}$ is primitive, $L_{+,2} = \mathbb{Z}_2\lambda_0$, and with $e_1 = \operatorname{diag}(1,-1)$, $e_2 = \begin{pmatrix} 0 & -d/4 \\ 1 & 0 \end{pmatrix}$:*

1. *$L_{-,2} = \langle -1 \rangle \perp \langle -|d|/4 \rangle$. If $d_0$ is odd this is the plane $\langle -1 \rangle \perp \langle -4^{k-1}c \rangle$, $c \equiv 3 \bmod 4$, of Lemma 18, with the same $2^k$ integral cosets $(0, t/2^{k-1})$ and $(1/2, t/2^k)$ and the same series and $\kappa_\nu^-(0)$ (Lemma 20); if $d_0$ is even it is $\langle -1 \rangle \perp \langle -4^k c \rangle$ with $c \equiv 1 \bmod 4$ or $\langle -1 \rangle \perp \langle -2^{2k+1}c \rangle$, anisotropic, with integral cosets $(0, r/2^k)$, $\rho = \operatorname{ord}_2 r$, series $\alpha_j = 2^{\lfloor j/2 \rfloor}$ for $j \leq 2\rho + 1$ and 0 beyond (zero coset: up to $j \leq 2k+1$, then $2^k$), and $\kappa_{\nu_r}^-(0) = -2(2^{\rho+1} - 1)/2^k \cdot \log 2$, the formula of Lemma 21(4) at $q = 2$.*
2. *$L_2^\vee = L_2 + \mathbb{Z}_2 e_1/2$ meets $L_{-,2} \otimes \mathbb{Q}$ in $L_{-,2} + \mathbb{Z}_2 e_1/2$, and $e_1/2$ has norm $-\frac{1}{4}$: no nonzero integral coset of the plane lies in $L_2^\vee$. The cosets of the second kind $(1/2, t/2^k)$ pair with $x$ such that $x + \nu \in e_1/2 + L_2$, i.e. over the nonzero class of $(L^\vee/L)_2 \cong \mathbb{Z}/2$, at the exponents $Q(x) \equiv \frac{3}{4} \bmod \mathbb{Z}$; all others pair over the trivial class at 2.*

*Proof.* $R_f \otimes \mathbb{Z}_2 = \mathbb{Z}_2 + 2^k\mathcal{O}_{k,2} = \mathbb{Z}_2[2^{k-1}\sqrt{d_0}] = \mathbb{Z}_2[\lambda/2]$ whether $d_0$ is odd or even, so the embedding is optimal iff $\mathbb{Z}_2^2$ is a module over $\mathbb{Z}_2[\lambda/2]$ with that order as its full ring of multipliers; a lattice in a quadratic field whose multiplier ring is a given order is an invertible, hence principal, ideal of it, so $\mathbb{Z}_2^2 \cong \mathbb{Z}_2[\lambda/2]$ and in the basis $1, \lambda/2$ the element $\lambda/2$ acts as $\begin{pmatrix} 0 & (\lambda/2)^2 \\ 1 & 0 \end{pmatrix}$ with $(\lambda/2)^2 = d/4$. With $Q(a,b,c) = -a^2 - bc$, $\langle x, \lambda_0 \rangle = -(\beta + \gamma d/4)$ gives the

basis $e_1, e_2$ of $L_{-,2}$ and $Q(e_1) = -1$, $Q(e_2) = d/4$, $\langle e_1, e_2\rangle = 0$. For odd $d_0$, $|d|/4 = 4^{k-1}|d_0|f'^2$ with $|d_0| \equiv 3 \bmod 4$; for even $d_0 = 4m$, $|d|/4 = 4^k|m|f'^2$ with $|m| \equiv 1 \bmod 4$ or even, and $-c$ is then not a square mod 4, resp. the exponent is odd, so the counts of Lemma 20 stop after $n = 1$ exactly as in the ramified case of Lemma 21. The Gram matrix of $\langle\cdot,\cdot\rangle$ on $L_2$ in the basis $e_1, e_{12}, e_{21}$ is $(-2) \oplus \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$, so $L_2^\vee = L_2 + \mathbb{Z}_2 e_1/2$; a vector $\nu = se_1/2 + te_2/(|d|/2)$ has $\langle \nu, e_{21}\rangle = -t/2$ and $\langle \nu, e_{12}\rangle = -2t/|d|$, integral only for $t \equiv 0$. For (2), $x = t\lambda_0 = (0, td/4, t)$ and $x + \nu \in L_2^\vee$ iff $t + 2t'/|d| \in \mathbb{Z}_2$ (then the $\beta$-coordinate is integral automatically), for either value of $s$; when $s = 1$ the sum lies in $e_1/2 + L_2$, and $Q(x) = t^2|d|/4 \equiv -\frac{1}{4} \bmod \mathbb{Z}$ there. □

**Lemma 23** (a level prime dividing both the conductor and $d_0$)**.** *Let $p \,\|\, N$, $p \mid d_0$ and $p^k \,\|\, f$ with $k \geq 1$. After the conjugations of Lemma 17 (and, for $p = 2$, of Lemma 18), $\lambda = \begin{pmatrix} 0 & b \\ p^2c' & 0 \end{pmatrix}$ with $b$ a unit and $bc' = d/p^2$ for odd $p$, and $\lambda = \begin{pmatrix} 0 & 2b' \\ 4c' & 0 \end{pmatrix}$ with $b'$ odd, $c'$ even and $b'c' = d/8$ for $p = 2$. Then $L_{-,p} = \langle -1\rangle \perp \langle -|d|/p^2\rangle$ for odd $p$ and $\langle -1\rangle \perp \langle -|d|/4\rangle$ for $p = 2$: the anisotropic planes of Lemmas 21(4) and 22(1), with their integral cosets $(0, r/p^k)$, $\rho = \operatorname{ord}_p r$, their series, and*

$$\kappa^-_{\nu_r}(0) = -\frac{2(p^{\rho+1} - 1)}{(p-1)p^k} \log p.$$

*The index $[L_p : L_{+,p} \oplus L_{-,p}]$ is $p^{2k}$, and the cosets of order $p$ ($\rho = k - 1$), $p - 1$ of them for odd $p$ and one for $p = 2$, lie in $L_p^\vee$, over nonzero isotropic classes of $L^\vee/L$ supported at $p$. So, unlike the fundamental ramified case (Remark 11), the constant term enters: the $x = 0$ pairs contribute $\frac{p^k-1}{p^k}\,\frac{1}{2}c_\eta(0)\log p$ per point, and the $x \neq 0$ pairs the poles of $f$ at the CM points of $d/p^2, \ldots$ as before.*

*Proof.* The derivation of Lemma 17(1) used only $p^2 \mid Q(\lambda)$ and optimality, so it applies, with $a'^2 + bc' = d/p^2$ now divisible by $p$; normalising $a' = 0$ the plane is $-\alpha^2 - (p^2c'/b)\beta^2 = -\alpha^2 - (|d|/p^2)(1/b^2)\beta^2$ with $\operatorname{ord}_p(|d|/p^2) = 2k-1$ odd. For $p = 2$ the derivation of Lemma 18(1) gives $a'$ even (as $a'^2 + 2b'c' = -|d|/4 \equiv 0 \bmod 4$), $b'$ odd and $c'$ even, and the conjugation by $\begin{pmatrix} 1 & 0 \\ 2t & 1 \end{pmatrix}$ shifts $a'$ by $2tb'$, so $a' = 0$; the plane is $-\alpha^2 + (2c'/b')\beta^2$ with $2c'/b' = -|d|/(4b'^2) = -4^k|m|f'^2/b'^2$, $d_0 = 4m$. The series and $\kappa^-$ are those computed in Lemmas 21(4) and 22(1), which depend only on the plane. The dual pairing is $\langle e_2, x\rangle = -p\gamma + (p^2c'/b)\beta$ (resp. $-2\gamma + (2c'/b')\beta$), so $e_2/p^m \in L_p^\vee$ iff $m \leq 1$, and $Q(e_2/p) = -|d|/p^4 \in \mathbb{Z}_p$ (resp. $Q(e_2/2) = -|d|/16$) is integral since $k \geq 1$; its class in $(L^\vee/L)_p$ is $e_{12}/p$ (resp. the class of $e_{12}/2$), a nonzero isotropic vector. The index is the valuation of $\det(\lambda, e_1, e_2) = 2pc'$ (resp. of the determinant ratio $4^{2k}$ as in Lemma 18). The weight of the $x = 0$ pairs is $-\frac{1}{4} \cdot (p-1) \cdot \kappa^-_\nu(0) \cdot c_\eta(0)$ at $\rho = k - 1$ for odd $p$, and $-\frac{1}{4}\kappa^-_\nu(0)c_\eta(0)$ for $p = 2$ where $p - 1 = 1$. □

So at every prime of the conductor the term is determined, by Lemmas 17, 18, 21, 22 and 23.

Check. One published value reaches this case: on $X_0^{21}(2)$, [5] give $s(-16) = 1$, with $-16 = 2^2 \cdot (-4)$ and 2 the level prime dividing $d_0$. There the term is $-4, -6, -8, -4\log 2$ on four of the covers and 0 on the Hauptmodul forms (for them the $x = 0$ pair, with $\frac{1}{2}c_\eta(0)$, is cancelled by the pole of $f$ at $\tau_{-4}$ through $x = \pm\lambda_0/2$), so the Hauptmodul row cannot see it; but every form is a function of $s$ with known divisor, supported at $\tau_{-4}$ (the pole, where $s = \infty$), $\tau_{-7}$, $\tau_{-28}$, $\tau_{-84}$ and $\tau_{-168}$, all with published $s$. Fitting each form's constant at one fundamental

point and checking it at ten to thirteen others, the value at $-16$ forced by the divisor agrees with the pipeline's for all nine forms; without the term the four covers are off by $2^4, 2^6, 2^8, 2^4$ (`level-p2/ramcheck.py`).

Check. Here the published tables do reach the case, once the points are forced in: on the odd-level bases $X_0^{35}(1)$, $X_0^{39}(1)$, $X_0^{51}(1)$, $X_0^{55}(1)$ and $X_0^{57}(1)$, [5] list $s$ at $d = -28$ and $-60$ (2 split in $k$), $-12$ (2 inert), and $-16 = 2^2 \cdot (-4)$ ($d_0$ even), and at $-315 = 3^2 \cdot (-35)$ on $X_0^{35}(1)$ ($3 \mid d_0$). The terms are $\log 2$ on the Hauptmodul form at the split points, $-\log 2$ at $-12$ and $-16$ (and $-2, -4, -6, -8 \log 2$ on the covers), $\log 3$ at $-315$, and with them the pipeline reproduces every one of these eight published values, while every other row of those tables is unchanged (`tests/_offline/GuoYang_*.m`, which now force these points). Without a term at 2 the Hauptmodul value is off by a factor 2 at each of them.

**Remark 24** (the $m > 0$ terms at non-fundamental $d$)**.** For $m > 0$ the preprint's formula (its eq. (euler)), after [4],

$$A_\mu(s, m, v) = -\frac{2\pi}{L(s+1, \chi)} \prod_{p \in S} \frac{W_{m,p}(s, \varphi_\mu)}{1 - \chi(p) p^{-1-s}}, \qquad S = \{p : \ p \mid \operatorname{disc} L_- \text{ or } p \mid m\},$$

is an Euler-product identity: away from $S$ the plane is unimodular and $W_{m,p} = 1 - \chi(p)p^{-1-s}$ with $\chi$ the character of the quadratic space $L_- \otimes \mathbb{Q} \cong (k, -cN_{k/\mathbb{Q}})$, which is $\chi_{d_0}$ for every $d$ with the same fundamental discriminant. Nothing in the derivation of Theorem B uses that the order is maximal, and the formula at conductor $f$ is this one, evaluated on the plane of the order. We make the two points where $f$ could have entered explicit.

*The local factors.* For a coset $\mu$ of $L_-$ and $m > 0$ let $\alpha_k = p^{-k} \#\{x \in (\mu + L_{-,p})/p^k L_{-,p} : Q(x) \equiv m \bmod p^k\}$ and $W^{\text{count}} = (1 - X) \sum_k \alpha_k X^k$; at a unimodular prime with $p \nmid m$ this is $1 - \chi(p)X/p$. The polynomials the routine evaluates ([3, Thm. 4.3, 4.4] on the actual $L_{-,p}$, times $|\det S|_p^{1/2}$) coincide with $W^{\text{count}}$ at every $(d, p, m)$ tried on $X_0^{15}(2)$: $d = -15, -60, -240, -28, -12, -48$, $p = 2$ (the conductor planes) and $p = 7$ (unimodular, split or inert), $m \le 6$ (`level-p2/wnorm.m`). A prime of the conductor divides $\operatorname{disc} L_-$ (Lemmas 17, 18, 21, 22), so it lies in $S$ and its factor is one of these polynomials, never an unramified Euler factor.

*The global factor.* The routine evaluates, per CM point and up to the $-\log p'$ at the vanishing place,

$$\kappa_\mu^-(m) = \frac{w_0 \sqrt{|d_0|}}{h_0} \cdot \frac{\left[\prod_{p \in S} W_{m,p}(s, \varphi_\mu)\right]'_{s=0}}{\prod_{p \in S} (1 - \chi_{d_0}(p)/p)},$$

which is the identity above at $s = 0$: $L(1, \chi_{d_0})^{-1} = w_0 \sqrt{|d_0|}/(2\pi h_0)$ by the class number formula of the field, and the Euler correction at every prime of $S$, those of the conductor included (in the code $\sqrt{|d|}/f = \sqrt{|d_0|}$, and the Kronecker character of $d$, which Magma returns as the primitive character of conductor $|d_0|$; `level-p2/kronprod.m` lists the factors at $d = -60, -240, -28, -48, -12$ and at each $p \mid f$ the factor is $1 - \chi_{d_0}(p)/p$). This is what the values confirm: the 46 values on $X_0^{15}(2)$ and the eleven forced Guo–Yang points are exact with it. The earlier code had the order's class number $h(R_f)$ in place of $h_0$, hence was off by $h(R_f)/h_0$, visible only at conductor 4; that amounts to $L(1, \chi_d)$ of the imprimitive character in place of $L(1, \chi_{d_0})$, i.e. to the correction at $p \mid f$ applied twice. (An earlier version of this remark read the code's character as the imprimitive one and concluded that Theorem B at conductor $f$ carried an extra factor $\prod_{p \mid f}(1 - \chi_{d_0}(p)/p)$; it does not.)

*Why the constant of Theorem B is unchanged.* In Schofer's argument the CM cycle is the orbit $Z(d) = T(\mathbb{Q}) \backslash T(\mathbb{A}_f) / K_f$, $T$ the torus of $k^\times$ and $K_f$ the stabiliser of $\widehat{L_-}$, the image of $\widehat{R}_f^\times$.

As $T$ is commutative every point of the orbit has the same stabiliser $R_f^\times/\{\pm1\}$, so under the Tamagawa measure, of total volume 2, every point carries the volume $2/|Z(d)|$ and the theta integral over $T(\mathbb{Q})\backslash T(\mathbb{A})$ is $(2/|Z(d)|)\sum_{z\in Z(d)}\theta(z)$, as for the maximal order; the Siegel–Weil formula then gives the Eisenstein series of the quadratic space $L_-\otimes\mathbb{Q}$ with the Schwartz function of the lattice $L_-$. The order enters only through $|Z(d)|$, which is the number of CM points of discriminant $d$ that Theorem B already counts, and through the local factors above. Independently, Siegel's genus-average identity for the definite plane,

$$\frac{\sum_{L'} r_{L'}(M)/|O(L')|}{\sum_{L'} 1/|O(L')|} = \frac{c_\infty}{\sqrt{\det S}}\cdot\frac{\prod_{p\in S}\alpha_p(M)}{L(1,\chi_{d_0})\prod_{p\in S}(1-\chi_{d_0}(p)/p)},$$

with $\alpha_p$ the counting densities, holds with one universal constant at $d=-15,-60,-240,-48$ on $X_0^{15}(2)$ — the corrections at the conductor prime included — and fails without them (`level-p2/siegel.m`): the global factor of $E(\tau,0,\varphi_{L_-})$ does not depend on $f$, and $\kappa_\mu^-(m)$ is the same Euler product with one local factor differentiated.

**Remark 25** (a point on the divisor, and the value of a quotient there)**.** Theorem B is stated off the divisor of $\Psi$; at a point $\tau_d$ on it a single value is infinite, but a quotient of two forms with the same order there is finite, and [5, Table 45] gives one: on $X_0^{15}(2)$ every form has a pole at $\tau_{-12}$, and $F=f_{-2}/f_{-1}=(s-2)/(20s)$ has $F(-12)=1/20$. In the formula the point shows up as pairs $(x,\nu=0)$ with $Q(x)=m$ and $c_{[x]}(-m)\neq0$: for $f_{-2}$ the vectors $x=\pm\lambda_0$ with $m=3$ (the pole of $f$ at $\infty$), for $f_{-1}$ the vectors $x=\pm\lambda_0/2$ with $m=\frac34$ (the cusp-0 coefficient at $q^{-3/4}$). The routine used to treat these two kinds by two different rules taken from Yang's code, and gave $2^{2/3}/5$. Dropping both gives $1/5$: the pairs then cancel in the quotient, and what is missing is the constant the regularised Green function leaves at a singular pair, $-\log m$ per unit coefficient (Kudla's $\beta(2\pi vR)=-\gamma-\log(2\pi vR)+\dots$ with $R$ proportional to $m$ at the point), which differs between the two forms by $\log(3/\frac34)=2\log2$. With that single rule in place of the two heuristics, $F(-12)=1/20$ and also $f_{10}/f_{-1}^3=9/(2^{16}\cdot5)$, the value the divisors force for a quotient with three poles against three; the constant that does not depend on $m$ cancels in every quotient and is invisible in a single value, which is infinite anyway. Off the divisor no such pair occurs at any evaluated point, so nothing else changes.

Both quotients on $X_0^{15}(2)$ see the same gap, $\log3-\log\frac34=2\log2$. A second, independent gap is on $X_0^{21}(2)$ at $\tau_{-4}$, the pole of its Hauptmodul $s$ (the table of CM values of $X_0^{21}(2)$ in [5], $s(-4)=\infty$), where every form has a pole and the singular pairs are $x=\pm\lambda_0$ ($m=1$), $x=\pm3\lambda_0$ ($m=9$, the pole of $f_{13}$ at $q^{-9}$) and $x=\pm\lambda_0/2$ ($m=\frac14$). With $C_k$ read at one table point off each form's divisor and checked at the others, the 36 quotients $f_a^{\deg b}/f_b^{\deg a}$ of the nine forms (the four with half-integral principal parts doubled) all equal $C_a^{\deg b}/C_b^{\deg a}$ at $\tau_{-4}$, those through $f_{13}$ testing the gap $\log9-\log1=2\log3$ (`tests/DivisorPoint_21_2.m`).

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